

A hybrid clustering–optimization framework for continuous hub location in drone delivery networks

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ABSTRACT

This study aims to develop a scalable methodology for identifying optimal locations of micro-hubs and intermediate hubs within an intermodal drone delivery network. It addresses the need for novel facility location models tailored to the unique constraints of drone logistics, especially in continuous geographic spaces without predefined candidate sites. The research adopts a hybrid approach combining density-based clustering (DBSCAN) with an uncapacitated facility location model. DBSCAN is employed as a preprocessing tool to identify initial micro-hub candidate locations based on customer distribution. These are refined by a mathematical optimization model that assigns customers to the nearest feasible facility within drone range while penalizing unassigned customers. To quantify the impact of spatial discretization, a discretization loss analysis is conducted, assessing the deviation between continuous customer locations and centroid-based candidate hubs. For customers beyond drone range, intermediate hubs are introduced heuristically at fixed intervals, while a GIS-based relocation step ensures all facilities are placed in viable land areas, avoiding restricted or unusable zones, with displacement statistics reported to quantify real-world feasibility impacts. The methodology is applied in a case study involving two logistics companies in Germany. Results show that the proposed approach ensures complete customer coverage via drones, both with and without intermediate hubs, while maintaining operational feasibility and reducing complexity. Benchmark comparisons against optimal formulations, range sensitivity analysis, and scalability experiments demonstrate that the framework achieves near-optimal performance while scaling efficiently to larger urban instances. A final simulation confirms the efficiency and practicality of the designed delivery network. This work contributes a scalable decision-support tool for drone logistics planning, bridging clustering heuristics, mathematical optimization, and feasibility validation, and providing a foundation for future extensions incorporating capacity, routing, and sustainability considerations.

1. Introduction

The rapid advancement of drone technology has transformed logistics, particularly in last-mile delivery. Traditional ground-based methods face congestion, high costs, and environmental concerns, necessitating alternative logistics models [1,2]. Drone-based intermodal networks, supported by micro-hubs and intermediate hubs, offer a promising solution while addressing constraints such as limited battery range, payload, and regulatory restrictions [3]. However, these advantages can only be realized if the underlying logistics network is designed to explicitly account for drone endurance, charging opportunities, and geographic feasibility.

Designing such networks requires solving facility location problems (FLPs), which determine optimal hub placements to minimize cost or

distance while ensuring service coverage [4,5]. Since this study seeks hub locations without capacity restrictions and without predefined candidate sites, it is framed as an uncapacitated continuous-space FLP [6–8]. While continuous formulations provide modeling flexibility, they quickly become computationally intractable for large-scale, real-world instances. To balance flexibility with tractability, we adopt a hybrid approach that combines Density-Based Spatial Clustering of Applications with Noise (DBSCAN) with continuous optimization, enabling scalable candidate generation while retaining spatial adaptability.

Most drone logistics research has emphasized routing, scheduling, and operational efficiency rather than strategic facility location planning [9,10]. Existing studies on drone recharging infrastructure typically assume predefined candidate sites or impose restrictive grid

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discretizations, limiting their applicability in unconstrained geographic settings. This study addresses these gaps by proposing an uncapacitated, continuous-space facility location framework that determines both micro-hub and intermediate (relay) hub locations dynamically, without relying on predefined sites.

The framework is guided by three research questions:

1. How can uncapacitated facility location problems be adapted to intermodal drone delivery networks operating in continuous geographic space without predefined sites?
2. How can data-driven clustering be used to generate scalable micro-hub candidates while mitigating discretization loss?
3. How can drone range limitations be systematically handled to guarantee full customer coverage under realistic endurance assumptions?

This work contributes by proposing a hybrid clustering–optimization framework for strategic network design in drone logistics. The key contributions are as follows:

- *Hybrid clustering–optimization framework:* We integrate DBSCAN clustering with a continuous facility location model to identify micro-hub locations without predefined candidate sites, enabling scalability to large geographic regions.
- *Systematic outlier-handling:* We introduce a systematic interval-based heuristic that distinguishes between one-way delivery range (R) and inter-facility endurance ($2R$), allowing intermediate relay hubs to extend operational reach while preserving delivery feasibility at customer locations.
- *Discretization-aware and geographically feasible design:* We quantify and mitigate discretization effects arising from centroid-based clustering and apply a GIS-based snapping procedure to relocate hubs onto feasible land parcels, avoiding restricted or unusable areas.
- *Empirical validation, sensitivity, and scalability analysis:* The framework is evaluated on real-world datasets from two logistics companies in Germany. Sensitivity and scalability experiments demonstrate robustness to drone range variation and computational tractability as instance size increases.

Scope and terminology: This study examines the strategic network design of a hybrid truck–drone delivery system. The focus lies on determining feasible facility locations rather than on operational routing or scheduling decisions. In this framework, the warehouse serves as the primary supply node from which trucks transport goods to micro-hubs, while micro-hubs function as decentralized facilities enabling last-mile drone operations. Drones may deliver products directly from the warehouse or from a micro-hub, and when necessary, may traverse one or more intermediate (relay) hubs; charging-only facilities used to extend their operational range before reaching customers. The system is described as intermodal because trucks operate in the first echelon (warehouse to micro-hubs), whereas drones operate in the second echelon (warehouse or micro-hub to customers, with optional relay support).

The remainder of the paper is organized as follows. Section 2 reviews the related literature, and Section 3 outlines the methodological foundations of the proposed framework. Section 4 presents the problem formulation, while Section 5 reports the case study application and experimental results. Section 6 discusses the findings and their implications, and Section 7 concludes with a summary and directions for future research.

2. Literature review

This section reviews relevant literature in three areas central to this study: facility location problems, clustering approaches for demand aggregation, and optimization of intermodal delivery systems. Together, these topics provide the theoretical and methodological foundation for designing efficient, multi-tier drone delivery networks with micro-hubs and intermediate hubs.

2.1. Facility location problems (FLPs)

Facility location problems (FLPs) are central to operations research, concerned with determining optimal facility placements and demand assignments to minimize costs or maximize service performance [11, 12]. Classical foundations include the p -median and p -center problems [13], while early work highlighted the inherent complexity of location–allocation formulations [14]. These frameworks have evolved into a broad family of models addressing diverse logistical and strategic challenges [6].

FLPs can be classified into discrete and continuous formulations [15]. Discrete models restrict facilities to a finite set of candidate sites and include uncapacitated and capacitated FLPs, p -median/ p -center problems, and covering-based formulations such as location set covering problem and the maximal covering location problem [16–18]. Continuous models allow facilities to be sited anywhere in a feasible region [14,19], providing flexibility but requiring geometric or iterative optimization techniques due to NP-hardness in multisource problems [8,20]. Variants include continuous p -median/ p -center, Fermat-type, and obnoxious location models [21,22].

Research in FLPs has expanded along several complexity axes: richer constraints such as capacity, multi-echelon structures, modular openings, and equity considerations [12,23]; uncertainty and robustness with stochastic demand and costs [24]; network realism including travel-time, congestion, and multimodal transport [18,25]; and sustainability and resilience addressing energy, emissions, and network adaptability [26].

Solution methods range from exact approaches – branch-and-bound, branch-and-cut, Benders decomposition, and Lagrangian relaxation – for moderate discrete instances [6], to heuristics, metaheuristics, and matheuristics for larger or complex problems [27,28]. Increasingly, clustering-based preprocessing (e.g., k-means, DBSCAN) reduces problem dimensionality by grouping demand points and narrowing candidate facility sets [29,30].

While discrete models are tractable and continuous models offer flexibility, modern applications such as drone delivery require hybrid approaches. Range limitations and multi-tier network considerations necessitate integrating continuous optimization with clustering heuristics to identify feasible and efficient facility placements [31].

2.2. DBSCAN clustering approaches in the context of FLP

Clustering methods, such as DBSCAN, provide heuristic solutions to FLPs, particularly when large datasets make exact optimization computationally prohibitive [32]. By grouping demand points based on spatial proximity, DBSCAN helps identify potential facility locations while detecting clusters of varying shapes and sizes as well as outliers. However, clustering alone does not guarantee network-wide cost minimization, as it cannot account for interactions across different tiers of a logistics network.

In continuous facility location problems (CFLPs), DBSCAN can be combined with mathematical optimization to reduce computational complexity [33]. Here, customer locations are first clustered, allowing optimization models to determine facility placement for entire clusters rather than individual points. This hybrid approach leverages DBSCAN's ability to handle irregularly shaped clusters and outliers while benefiting from the precision of mathematical optimization. Its effectiveness depends on appropriate parameter tuning, such as the distance threshold (ϵ) and minimum points per cluster [34], and it may face challenges with highly variable demand patterns [35].

DBSCAN-based methods are computationally efficient and suitable for large-scale, dense urban problems. In contrast, hierarchical optimization models offer greater precision and flexibility to capture system-wide interactions and diverse spatial scenarios, but at higher computational cost [36–38]. Both approaches provide complementary

strengths, motivating hybrid strategies that balance efficiency and solution quality.

Several studies illustrate the potential of clustering in FLPs. Lin et al. [39] used hierarchical clustering on JD Logistics data, achieving faster computation and improved performance compared to traditional methods. Alvarez-Urbe et al. [40] combined Agglomerative Nesting (AGNES) with K-medoids to identify micro-hub locations for urban newspaper distribution. Wu et al. [41] proposed a two-stage model integrating an improved clustering algorithm with the center of gravity method, while Sharma et al. [42] combined DBSCAN with affinity propagation for two-stage facility allocation. Similarly, Sharma et al. [43] integrated k-means with mixed-integer programming to optimize facility placement in wireless networks and ATMs. Unlike k-means, DBSCAN does not require a predefined number of clusters and accommodates dense areas and outliers. Building on these concepts, the present study integrates DBSCAN clustering with mathematical optimization for continuous-space facility location problems, enabling flexible facility placement while considering drone range constraints.

2.3. Optimization of intermodal delivery systems

Intermodal delivery systems leverage multiple transportation modes such as trucks, trains, ships, and drones to optimize cost, speed, and environmental impact across varying distances and geographies. Drones have gained attention for last-mile delivery due to their ability to bypass traffic congestion, reduce delivery times, and reach hard-to-access areas [44,45]. In drone-based systems, *micro-hubs* are small, strategically placed facilities where drones can load, unload, and recharge or swap batteries. Acting as intermediaries between central warehouses and customers, micro-hubs shorten drone travel distances, enable faster deliveries, and increase network flexibility [46]. Incorporating drones and micro-hubs improves supply chain efficiency and customer experience but also introduces challenges such as regulatory restrictions, technological barriers, and the need for advanced algorithms for routing, scheduling, and infrastructure placement [47,48].

Several studies have developed optimization models for such systems. Wang et al. [49] used fuzzy theory to identify optimal drone service facility locations in emergency services, balancing cost, efficiency, and response time. Hong et al. [50] developed a model for siting drone recharging stations, integrating shortest-path and coverage models to ensure feasible delivery networks. Shavarani et al. [51] optimized charging station placement from predefined candidate locations. While these studies focus on discrete candidate sites, our work determines possible locations for micro-hubs, charging stations, and launch facilities in continuous space, considering drone range limitations.

Few studies address continuous-space facility location. Meira et al. [29] reduced the continuous problem to approximate center sets, achieving theoretical guarantees without practical validation. Shavarani et al. [52] proposed a multi-level model incorporating recharging, warehouse, and transportation costs with fuzzy and queueing theory for uncertainty. Zhao et al. [15] introduced a mixed discrete-continuous model with zone-based cost variations, though irregular zones remain challenging. Gao et al. [7] used weighted medians in continuous multi-facility problems.

Despite the growing body of work on drone-assisted logistics and facility location, several methodological gaps remain. Many studies assume a discrete and predefined set of candidate hub locations, limiting flexibility in emerging markets where infrastructure is not yet established. Continuous facility location models address this limitation but often suffer from scalability issues and typically do not incorporate realistic service range constraints or geographic feasibility. Clustering-based approaches have been proposed to reduce complexity; however, they usually rely on partitioning methods such as k-means, require the number of clusters to be specified a priori, or do not explicitly address isolated demand points.

In contrast, the present study integrates density-based clustering with continuous-space facility location by using DBSCAN to generate candidate micro-hub locations while explicitly preserving outlier customers for further treatment. These outliers are subsequently served through a feasibility-driven intermediate hub heuristic, ensuring full coverage under strict drone range constraints. The framework is further distinguished by incorporating GIS-based feasibility checks that translate continuous solutions into deployable locations. Together, these elements form a unified methodology that bridges continuous modeling, discrete optimization, and practical feasibility. Table 1 compares some representative previous works to the current work.

In summary, optimizing intermodal delivery systems with drones and micro-hubs requires balancing efficiency, adaptability, and scalability. By combining clustering methods with mathematical optimization, this research addresses practical challenges in locating micro-hubs and intermediate hubs, contributing to sustainable and efficient multi-tier drone delivery networks.

3. Methodological background

This section outlines the methodological foundations of the proposed framework. The approach combines continuous facility location modeling, density-based clustering, and heuristic feasibility recovery to support strategic design of intermodal drone delivery networks. The framework is intended for strategic planning and network configuration; it does not model detailed operational routing, scheduling, or real-time dispatching.

3.1. Continuous facility location problems for drone logistics

Facility location problems (FLPs) are a central topic in operations research and logistics planning, aiming to identify optimal facility locations that serve spatially distributed demand at minimum cost or distance [18,56]. In classical discrete FLPs, facilities are selected from a predefined candidate set. In contrast, the continuous facility location problem (CFLP) allows facilities to be placed anywhere in a continuous plane, making it particularly suitable for drone logistics, where aerial vehicles are not constrained by road infrastructure and can theoretically operate from any feasible open location [57].

Let I be the set of demand points (customers), with each customer $i \in I$ located at coordinates (x_i, y_i) . We aim to find the coordinates (u_j, v_j) for a set of facilities J (e.g., micro-hubs). The distance between customer i and facility j is calculated using the Euclidean metric:

$$d_{ij} = \sqrt{(x_i - u_j)^2 + (y_i - v_j)^2}.$$

The classical CFLP objective is to minimize the sum of distances from each customer to their nearest assigned facility, formulated as the Weber problem or p -median problem in its multi-facility variant:

$$\min_{(u_j, v_j)} \sum_{i \in I} w_i \min_{j \in J} d_{ij}, \quad (1)$$

where w_i is a weight associated with customer i , often representing demand.

For drone-based delivery, this model must be augmented with operational constraints. A critical constraint is the maximum flight range of the drone, R . A customer i can only be feasibly served by a facility j if their distance does not exceed this range, i.e., $d_{ij} \leq R$. This constraint transforms the problem into a covering-type model, where the goal is to locate facilities such that all customers are within the service radius of at least one hub. Directly solving such a constrained CFLP in continuous space is computationally prohibitive for large-scale, real-world instances. The proposed framework therefore introduces a structured discretization via clustering, combined with downstream feasibility recovery mechanisms.

Table 1
Comparison with related facility location approaches in drone logistics.

Aspect	Discrete FLP [50,53]	Continuous CFLP [8]	Clustering- assisted FLP [54,55]	This study
Facility space	Discrete	Continuous	Discrete	Continuous → discrete
Clustering method	–	–	k-means/sampling	DBSCAN (density-based)
Outlier handling	Penalized/ignored	Penalized	Ignored	Relay-based feasibility
Range constraints	Simplified	Limited	Simplified	Explicit (R , $2R$)
GIS feasibility	No	No	No	Yes

3.2. DBSCAN clustering approach

Clustering analysis partitions data into groups (clusters) of similar points and is widely applied in machine learning, data mining, and logistics planning [58]. Among various clustering methods such as K-Means, Gaussian Mixtures, and Spectral Clustering, DBSCAN is particularly suitable for spatial logistics data, as it detects clusters of arbitrary shapes and identifies noise points (outliers) without requiring the number of clusters to be predefined [34,59,60].

DBSCAN relies on two parameters: ϵ (the neighborhood radius) and MinPts (the minimum number of points to form a dense region) [61]. For a point p , the ϵ -neighborhood is defined as:

$$N_{\epsilon}(p) = \{q \in D \mid \text{dist}(p, q) \leq \epsilon\},$$

where $\text{dist}(p, q)$ is the Euclidean distance. A point is a *core point* if $|N_{\epsilon}(p)| \geq \text{MinPts}$. Clusters are then formed by linking core points and their neighborhoods.

DBSCAN classifies points into three categories: (i) *core points*, which have sufficient neighbors to form clusters; (ii) *border points*, which fall within the neighborhood of core points but lack enough neighbors to form clusters themselves; and (iii) *noise points*, which are not part of any cluster [61]. The robustness of DBSCAN to noise and its flexibility in discovering clusters of varying density make it especially valuable for modeling real-world delivery demand.

Parameter selection

The performance of DBSCAN depends on selecting suitable values for ϵ (neighborhood radius) and MinPts (minimum points to form a cluster). If ϵ is too small, many points are labeled as noise, while overly large values may merge distinct clusters. A common approach is the *k-distance graph*, where the distance to each point's k th nearest neighbor is plotted, and the “elbow” indicates an appropriate ϵ [62,63]. MinPts is typically set as $D + 1$, where D is the dataset's dimensionality; for 2D spatial data, a typical starting value is MinPts = 3 [40].

In our study, the initial ϵ value is estimated using the k-distance method in relation to the maximum drone delivery range, and refined iteratively within the optimization framework. Cluster centroids then serve as candidate micro-hub locations, while noise points are interpreted as outlier customers for separate treatment.

The DBSCAN neighborhood radius ϵ is selected based on the operational range of the drone fleet. Specifically, ϵ is chosen such that customers within a radius corresponding to feasible last-mile drone delivery are grouped into the same cluster. This aligns the clustering process with the physical constraints of the delivery system. The minimum number of points parameter, *MinPts*, reflects the smallest customer density required to justify opening a micro-hub and is therefore set to a small integer value consistent with sparse-demand settings.

3.3. Integration for drone network design

The proposed framework integrates clustering, optimization, and heuristic feasibility recovery in a structured workflow. First, DBSCAN

is applied to customer coordinates to identify dense demand clusters and noise points. Second, cluster centroids are used as candidate micro-hub locations in a mixed-integer optimization model that assigns customers to facilities under drone range constraints. This step determines which micro-hubs are activated and identifies customers that remain infeasible under direct drone delivery.

Third, customers that cannot be served within range R from any warehouse or micro-hub are handled through an interval-based relay hub heuristic. The heuristic exploits the distinction between (i) a one-way delivery range R for terminal customer legs and (ii) an endurance of up to $2R$ for facility-to-facility flights where recharging is available. Intermediate (relay) hubs are placed along straight-line paths at fixed intervals to restore feasibility without modifying the original micro-hub structure.

Fourth, all candidate facilities (micro-hubs and relay hubs) are re-located to the nearest feasible land parcels using a GIS-based snapping procedure that avoids restricted, unusable, or unsuitable areas. Finally, a simulation step validates the resulting network design under operational constraints, ensuring that all customers can be served within drone endurance limits.

Fig. 1 summarizes the complete methodological workflow. By combining data-driven clustering, optimization-based selection, heuristic feasibility recovery, and geographic validation, the framework balances realism, scalability, and computational tractability, making it suitable for large-scale strategic planning of drone delivery networks.

4. Problem description and formulation

4.1. Introduction

This section presents the modeling framework for optimizing facility locations in a drone-based last-mile delivery system. We begin by describing the logistics network and delivery constraints, followed by a conceptual continuous facility location model that captures the spatial nature of the problem. Due to the computational complexity of solving the continuous model directly, we introduce a clustering-based discretization strategy using DBSCAN, which enables reformulation as a tractable integer programming model (generally, mixed-integer linear program (MILP)). The final model determines optimal micro-hub placements and customer assignments while minimizing total transportation distance under drone range constraints.

4.2. Problem description

We consider a three-tier logistics network consisting of a centrally located warehouse, a set of geographically distributed customers, and a set of micro-hubs that must be strategically established to serve last-mile demand. The modeled system is representative of small-package distribution suitable for drone delivery, such as pharmaceuticals, and spare part items that fall within typical payload and endurance limits of current delivery drones.

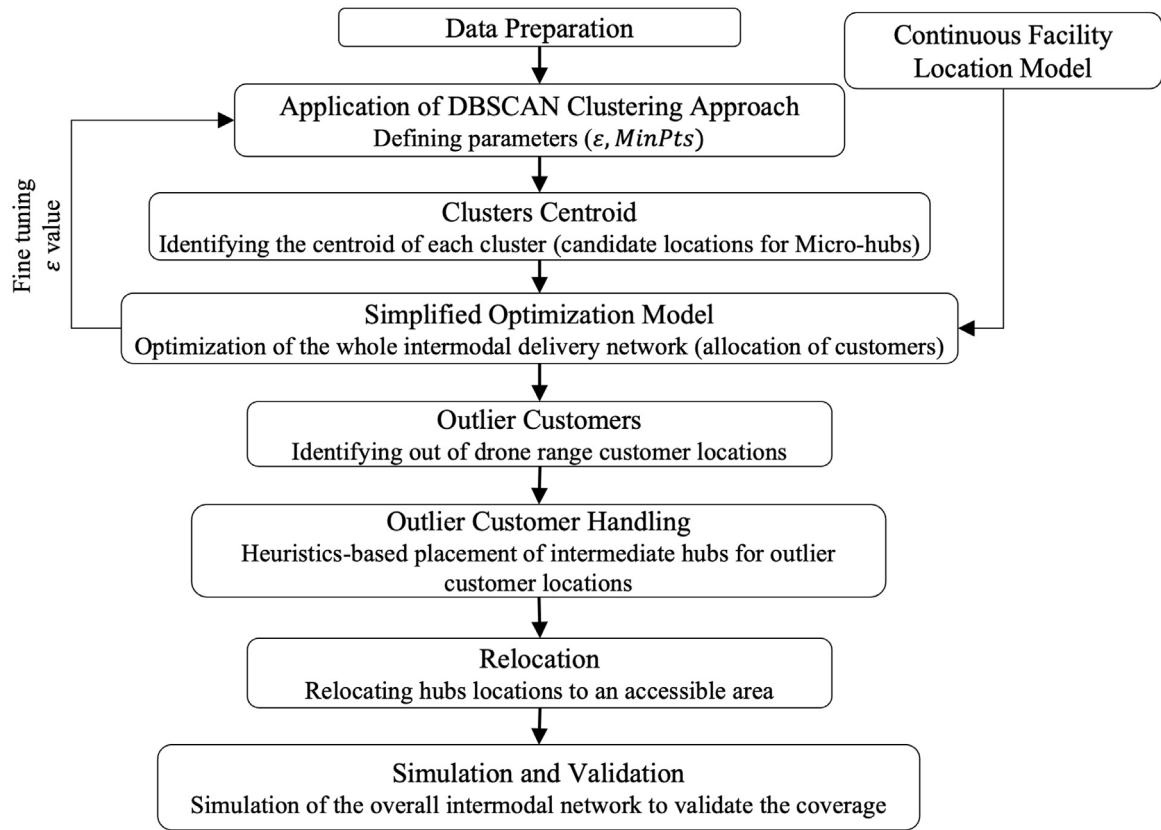


Fig. 1. Workflow of the proposed hybrid facility location framework integrating DBSCAN clustering, MILP-based selection, relay-based feasibility recovery, and GIS-based validation.

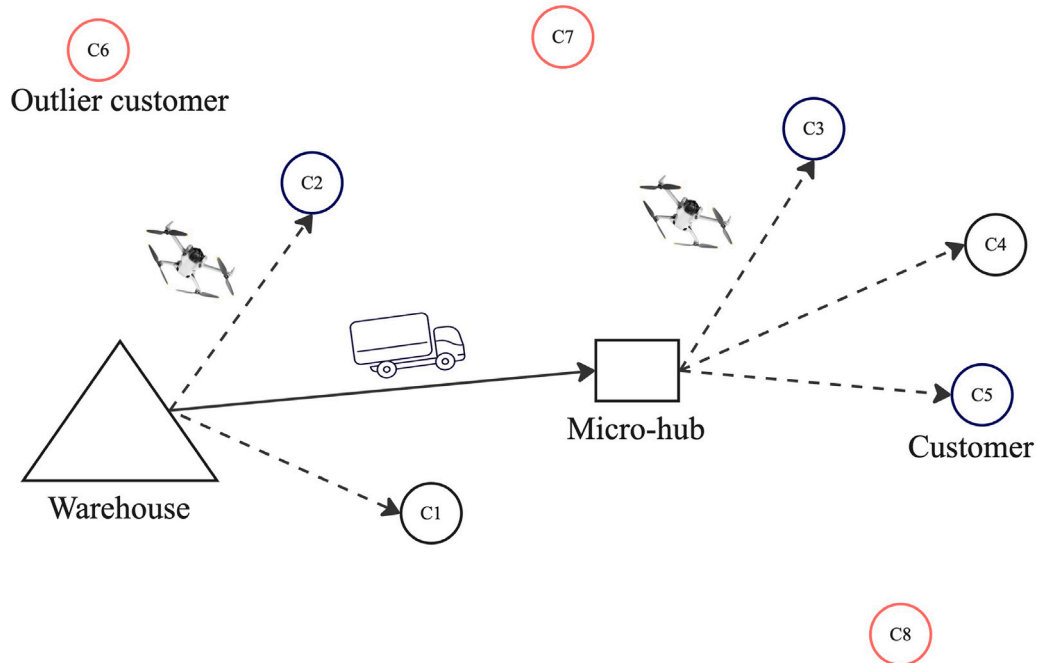


Fig. 2. Schematic of the three-tier logistics network, consisting of a central warehouse, potential micro-hub locations, and customer demand points. Out of drone range customer locations remain unassigned.

An illustrative example of the facility location problem under consideration is shown in Fig. 2. The figure depicts a representative drone-enabled small-package distribution network, where parcels are transported from the warehouse to micro-hubs by trucks and subsequently

delivered to customers by drones. Each customer must be assigned to a single facility: either a micro-hub or the warehouse provided the delivery distance does not exceed the drone's maximum range. If no feasible assignment exists, the customer may remain unserved, incurring a

penalty cost. This abstraction captures the key spatial and operational characteristics of emerging drone-supported last-mile logistics systems.

The facility location problem focuses on determining optimal placements for micro-hubs and assigning customers to these facilities in a way that minimizes the total transportation distance across the network. This total distance accounts for three key delivery legs: truck deliveries from the warehouse to micro-hubs, drone deliveries from micro-hubs to customers, and direct drone deliveries from the warehouse to customers. By strategically selecting micro-hub locations and assigning customers accordingly, the system aims to achieve an efficient and cost-effective logistics structure.

No upper bound is placed on the number of micro-hubs that can be established. However, their placement must ensure spatial feasibility and regulatory compliance.

4.3. Formulation

4.3.1. Conceptual continuous facility location model

We begin by conceptualizing the problem as a continuous facility location model. In this formulation, the number of micro-hubs is not fixed a priori, and their locations are continuous decision variables in $\mathbb{R}^2$.

Notation:

- W : Set of warehouses; warehouse $i \in W$ has fixed coordinates (x_i, y_i) .
- C : Set of customers, with customer $k \in C$ at coordinates (x_k, y_k) .
- M : An index set for potential micro-hubs to be located.
- $(u_j, v_j) \in \mathbb{R}^2$: Continuous coordinates of micro-hub $j \in M$ (decision variables).
- R : Maximum drone one-way range.
- $d(\cdot, \cdot)$: Euclidean distance metric.

The objective is to minimize the sum of total transportation distance. Each customer is assigned to the nearest open and feasible facility.

$$\min_{(u_j, v_j)_{j \in M}} \sum_{i \in W} \sum_{j \in M} d((x_i, y_i), (u_j, v_j)) + \sum_{k \in C} \min \left\{ \min_{i \in W} d((x_i, y_i), (x_k, y_k)), \min_{j \in M} d((u_j, v_j), (x_k, y_k)) \right\} \quad (2)$$

Feasibility Constraints:

$$d((u_j, v_j), (x_k, y_k)) \leq R, \quad \forall j \in M, k \in C$$

$$d((x_i, y_i), (x_k, y_k)) \leq R, \quad \forall i \in W, k \in C$$

In this model, customers are implicitly assigned to the nearest feasible facility either a warehouse or a micro-hub through the use of the *min* operator, which selects the shortest valid delivery path. The number and geographic placement of micro-hubs are treated as decision variables within a continuous spatial domain, allowing for flexible but computationally intensive optimization. The immense computational complexity of finding an exact solution motivates our approach of discretizing the search space using a data-driven clustering method.

4.3.2. Discretization via clustering

To reduce complexity, we discretize the continuous facility location problem by clustering customers into dense regions and extracting representative candidate hub locations. Following Chakrabarty et al. [55], such discretization methods yield scalable algorithms with provable approximation guarantees.

Specifically, we apply the density-based spatial clustering of applications with noise (DBSCAN) to customer geolocation data. DBSCAN identifies dense groups of customers without requiring the number of clusters to be predefined and designates isolated points as noise, thereby capturing realistic customer density. For each identified cluster, the geometric centroid is computed and treated as a candidate micro-hub location. This transformation converts the continuous-space model

Table 2

Notation used in the MILP formulation.

Sets	
C	Set of customers, indexed by k
M	Set of candidate hubs (cluster centroids), indexed by j
W	Warehouse location, indexed by i
Parameters	
d_{ij}	Distance from warehouse $i \in W$ to hub $j \in M$
d_{ik}	Distance from warehouse $i \in W$ to customer $k \in C$
d_{jk}	Distance from hub $j \in M$ to customer $k \in C$
R	Maximum drone delivery range
Ω	Penalty cost for each unassigned customer.
α	Truck distance weighting coefficient.
β	Drone distance weighting coefficient.
Decision Variables	
$h_j \in \{0, 1\}$	1 if hub j is opened, 0 otherwise
$a_{jk} \in \{0, 1\}$	1 if customer k is assigned to hub j
$b_{ik} \in \{0, 1\}$	1 if customer k is served directly by the warehouse
$e_{ij} \in \{0, 1\}$	1 if hub j is supplied by warehouse i
$p_k \in \{0, 1\}$	1 if customer k remains unassigned, 0 otherwise.

into a discrete MILP formulation, enabling exact optimization over binary decision variables. The procedure is summarized in Algorithm 1.

Algorithm 1 DBSCAN Clustering

- 1: **Input:** Customer coordinates (Longitude, Latitude)
- 2: **Parameters:** DBSCAN neighborhood radius ϵ , minimum samples m
- 3: Apply DBSCAN to customer coordinates with parameters (ϵ, m)
- 4: Assign each customer a cluster label; noise points are labeled -1
- 5: **for all** clusters $c \neq -1$ identified by DBSCAN **do**
- 6: Compute centroid of cluster c as the mean longitude and latitude of all customers in c
- 7: Mark centroid as a candidate micro-hub location
- 8: **end for**
- 9: Collect customers labeled -1 (noise) into a set of *outlier customers*
- 10: **Output:**
 - Cluster centroids $\rightarrow$ candidate micro-hub locations
 - Noise points $\rightarrow$ outlier customers (handled by intermediate hub heuristic)

4.3.3. Simplified MILP model

The sets, parameters, and decision variables are defined in Table 2. *Objective Function*

$$\text{Min } Z = \sum_{i \in W} \sum_{j \in M} \alpha \cdot d_{ij} \cdot e_{ij} + \sum_{k \in C} \beta \cdot \left(\sum_{i \in W} d_{ik} \cdot b_{ik} + \sum_{j \in M} d_{jk} \cdot a_{jk} \right) + \sum_{k \in C} \Omega \cdot p_k \quad (3)$$

Constraints

$$\sum_{j \in M} a_{jk} + \sum_{i \in W} b_{ik} + p_k = 1, \quad \forall k \in C \quad (4)$$

$$a_{jk} \leq h_j, \quad \forall j \in M, \forall k \in C \quad (5)$$

$$e_{ij} \leq h_j, \quad \forall i \in W, \forall j \in M \quad (6)$$

$$d_{jk} \cdot a_{jk} \leq R, \quad \forall j \in M, \forall k \in C \quad (7)$$

$$d_{ik} \cdot b_{ik} \leq R, \quad \forall i \in W, \forall k \in C \quad (8)$$

Distance Formulation in Discrete Space

Let:

- (x_i, y_i) : coordinates of warehouse $i \in W$
- (x_j, y_j) : coordinates of candidate micro-hub $j \in M$
- (x_k, y_k) : coordinates of customer $k \in C$

Then the Euclidean distances used in the MILP model are defined as:

$$d_{ij} = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}$$

$$d_{jk} = \sqrt{(x_k - x_j)^2 + (y_k - y_j)^2}$$

$$d_{ik} = \sqrt{(x_k - x_i)^2 + (y_k - y_i)^2}$$

The objective function in Eq. (3) minimizes the total transportation distance and penalties. The first term is the truck delivery cost from the warehouse to all established micro-hubs. The second and third terms represent the drone delivery distances from the warehouse and micro-hubs to customers, respectively. The final term penalizes unassigned customers.

Constraint (4) ensures that each customer is either assigned to a micro-hub, served from the warehouse, or left unassigned. Constraints (5) and (6) enforce that customers and warehouse can only be assigned to established micro-hubs, respectively. Constraints (7) and (8) enforce the drone range limit. This discrete formulation is a tractable MILP that can be solved using standard optimization solvers.

The coefficients α and β represent the relative importance of truck-based and drone-based travel, respectively. In the baseline case, we set $\alpha = \beta = 1$, thereby minimizing total traveled distance as a proxy for cost and travel time at the strategic planning level. This formulation allows alternative calibrations reflecting different cost structures, energy consumption, or environmental impacts to be incorporated directly.

4.3.4. Quantifying discretization effects

To quantify the loss introduced by centroid-based discretization, we conduct a local continuous refinement analysis for each activated micro-hub. Given the set of customers assigned to a hub by the MILP, a locally optimal continuous location is computed by minimizing the sum of Euclidean distances to those customers. The snap distance is considered as the Euclidean distance between the centroid-based hub location and this locally refined point. To assess operational relevance, snap distances are normalized by the drone range R used in Constraints (7)–(8). We further compare total drone travel distances before and after refinement to assess objective degradation due to discretization. For detail discretization information refer [Appendix A](#).

4.4. Feasibility-driven routing for outlier customers

While the MILP model effectively assigns clustered customers to nearby hubs, a subset of outlier customers may still lie beyond any feasible direct drone range. To guarantee service to these customers without violating endurance limits, we design a multi-leg routing strategy supported by intermediate hubs.

We assume that drones can cover a distance of up to $2R$ without recharging, reflecting their round-trip endurance before requiring a battery swap (recharge). Thus, on multileg routes, drones can travel up to $2R$ between two consecutive facilities (warehouse, micro-hub, or intermediate hub), recharge at the hub, and then continue to the next leg. However, since customers do not provide recharging facilities, the final leg from the last hub to the customer is restricted to at most R . This ensures that the drone can complete the delivery and return safely to the hub for recharging.

To operationalize this, we adopt an interval-based placement heuristic. For each outlier customer, the straight-line path from the nearest facility is divided into the minimum number of feasible segments. Intermediate hubs are positioned such that each facility-to-hub or hub-to-hub segment does not exceed $2R$, while the last hub-to-customer segment remains within R . For an outlier customer k at distance D_{fk} from its nearest facility f , feasibility requires the final delivery leg to be no longer than R , while intermediate legs may span up to $2R$. The minimum number of relay hubs required therefore satisfies

$$n_{hubs} = \max \left\{ 0, \left\lceil \frac{D_{fk} - R}{2R} \right\rceil \right\}.$$

The heuristic places relay hubs evenly along the straight-line segment connecting f and k . [Fig. 3](#) illustrates this routing scheme.

This heuristic is consistent with approaches in drone delivery literature where simplified routing ensures feasibility [64,65]. Our method offers a scalable solution for routing to outlier customers. The logic is outlined in [Algorithm 2](#).

Algorithm 2 Asymmetric Interval-Based Hub Placement for Outliers

```

1: for all outlier customer  $k$  do
2:   Find nearest facility  $f$  (warehouse or an opened micro-hub)
3:   Compute total distance  $D \leftarrow \text{Distance}(f, k)$ 
4:   if  $D > R$  then
5:     Let  $\vec{v}$  be the unit vector from  $f$  to  $k$ 
6:     Define the location of the last intermediate hub:  $h_{\text{last}} \leftarrow k - R \cdot \vec{v}$ 
7:     Compute remaining distance to cover:  $D' \leftarrow D - R$ 
8:     if  $D' > 0$  then
9:       Compute number of intermediate hubs needed:  $n_{\text{hubs}} \leftarrow \lceil D' / (2R) \rceil$ 
10:      Compute length of intermediate segments:  $L_{\text{inter}} \leftarrow D' / n_{\text{hubs}}$ 
11:      for  $i = 1$  to  $n_{\text{hubs}} - 1$  do
12:        Place intermediate hub  $h_i$  at position  $f + i \cdot L_{\text{inter}} \cdot \vec{v}$ 
13:      end for
14:      Place the last intermediate hub,  $h_{n_{\text{hubs}}}$ , at position  $h_{\text{last}}$ 
15:      Define route  $f \rightarrow h_1 \rightarrow \dots \rightarrow h_{n_{\text{hubs}}} \rightarrow k$ 
16:    else
17:       $\triangleright$  Customer is within direct range, this case is handled by the main model.
18:    end if
19:  end if
20: end for
```

While we implement the deterministic interval-based heuristic here for computational efficiency and scalability, its theoretical foundation is rooted in a network flow optimization model that explicitly selects intermediate hub locations and delivery paths. To validate the heuristic's performance, we compare its results against a MILP benchmark for smaller problem instances, confirming that the heuristic provides near-optimal path lengths without the prohibitive overhead of a full routing model. See [Appendix B](#) for more on benchmarking. We leave a full routing optimization and hub functionality (e.g., charging-only vs. package-handling) for future work.

The model addresses the strategic network design level, focusing on spatial hub placement under drone range constraints. Operational decisions such as routing, scheduling, and fleet sizing are intentionally not modeled. Travel distance serves as a proxy for operational effort and can be reweighted to reflect time, cost, or energy considerations. Micro-hub capacities are not explicitly constrained, as capacity decisions are typically adjustable at the operational level (e.g., fleet size or operating hours). Instead, feasibility is ensured through strict range constraints and assignment consistency. In practice, decision-makers may introduce conservative range buffers or validate selected configurations through subsequent routing or simulation analyses. The framework is thus intended as the strategic layer within a hierarchical planning process.

5. Case study

To validate the proposed framework, a case study was conducted using anonymized customer location data from two logistics companies operating in North Rhine-Westphalia, Germany, referred to as Company A and Company B. The primary reason for including two companies was to demonstrate the robustness and generalizability of the proposed framework across structurally different logistics networks. Company A represents a relatively small and Company B exhibits a larger with more geographically dispersed demand pattern. The study demonstrates the

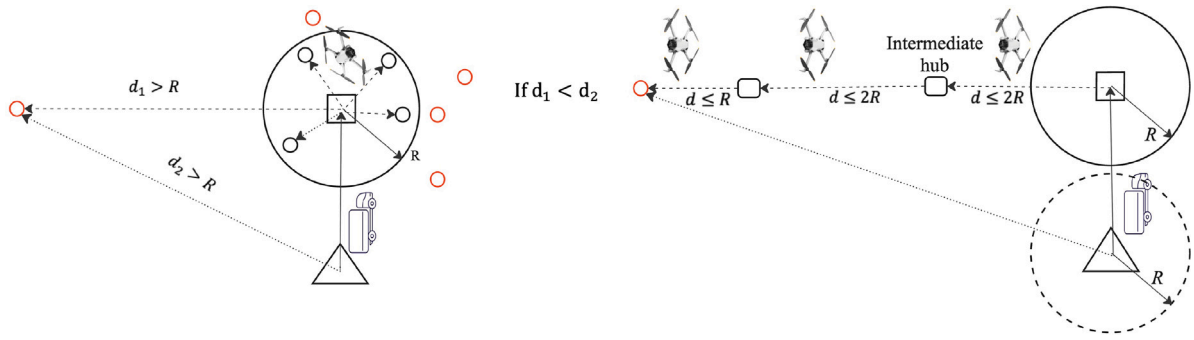


Fig. 3. Routing scheme for an outlier customer using intermediate hubs. When the distance from the nearest facility to the customer exceeds the drone range R , relay hubs are placed along the path to create a multi-leg journey where each segment is range-feasible.

Table 3
Iterative fine tuning of ϵ with the resulting clusters and customer assignments.

Iter.	ϵ	Company A			Company B		
		No. of clusters	Locations clustered	Outliers	No. of clusters	Locations clustered	Outliers
1	0.020	8	62	26	44	229	128
2	0.021	8	62	26	44	235	122
3	0.022	8	63	25	42	238	119
4	0.023	9	66	22	41	245	112
5	0.024	8	66	22	40	249	108
6	0.025	8	70	18	38	249	108
7	0.026	8	71	17	38	253	104
8	0.027	8	72	16	36	255	102
9	0.028	8	72	16	33	258	99
10	0.029	8	72	16	34	264	93
11	0.030	8	72	16	33	267	90
Total customers = 88				Total customers = 357			

effectiveness of our sequential methodology: identifying candidate hubs via DBSCAN clustering, optimizing the network with a MILP model, routing to outliers with a feasibility heuristic, and refining locations with a final snapping process.

5.1. DBSCAN clustering for candidate hub identification

Fig. 4 illustrates the spatial distribution of customer locations for both companies. For clustering, DBSCAN algorithm (see Section 4.3.2) is employed.

To determine appropriate ϵ values for DBSCAN, k-distance plots were generated (Fig. 5). The selection of ϵ value was guided by a two-stage procedure combining density diagnostics and operational objectives. Initially, the k-distance method was used to identify a plausible range for ϵ . Within this range, values of ϵ between 0.02 and 0.03 were evaluated in increments of 0.001 for both case companies. Guided by the k-distance plots and operational considerations, an iterative tuning process (Table 3) led to the selection of $\epsilon = 0.023$ for Company A and $\epsilon = 0.026$ for Company B, with MinPts set to 3 for both.

Applying DBSCAN with these parameters yielded 9 clusters for Company A and 38 for Company B, as shown in Fig. 6. Each cluster represents a dense customer region, and its geometric centroid was calculated to serve as a candidate micro-hub location for the subsequent optimization phase. The process also identified 22 customers from Company A and 104 from Company B as outliers due to their location in sparsely populated areas. These outliers are handled later in the routing stage.

The primary objective at the clustering stage was not to enforce drone delivery feasibility directly, but to obtain a compact and informative discretization of customer demand. Accordingly, customers classified as outliers at this stage are not interpreted as infeasible demand, but rather as inputs to a subsequent feasibility-recovery mechanism based on intermediate relay hubs. Specifically, ϵ was selected

to maximize the number of clusters while minimizing the number of outlier customers. This balance ensures that candidate micro-hub locations, defined as cluster centroids, are spatially representative and sufficiently granular for subsequent optimization. Using the selected ϵ values, DBSCAN clustering was integrated into the MILP formulation, where drone range constraints were explicitly enforced.

5.2. Optimization model results and delivery assignment

Using the cluster centroids as candidate sites, the optimization model (model in Section 4.3.3) was solved to configure the three-tier delivery network. The model selects an optimal subset of micro-hubs to activate and assigns customers to either a hub or the central warehouse, subject to an 8 km drone range limit.

The optimized network layouts are shown in Fig. 7. For Company A, the model activated 6 of the 9 candidate hubs, while 24 of the 38 candidate hubs were activated for Company B. The strategic placement of these hubs in high-density areas allows for efficient drone dispatch, covering numerous customers within a short radius. As a result, the majority of clustered customers were assigned to a micro-hub. A smaller number of customers, though clustered, were found to be more efficiently served directly from the warehouse.

The optimization model deliberately leaves a subset of customers unassigned when no direct drone-feasible facility exists within range R . These customers correspond primarily to spatial outliers and are subsequently handled through the relay-based feasibility restoration procedure described next. This group, comprises 6 customers for Company A and 42 for Company B. The summary of the result is shown in Table 4.

5.2.1. Truck-drone distance weights sensitivity analysis

To assess the robustness of the proposed framework, we conducted a sensitivity analysis with respect to the relative weighting of truck

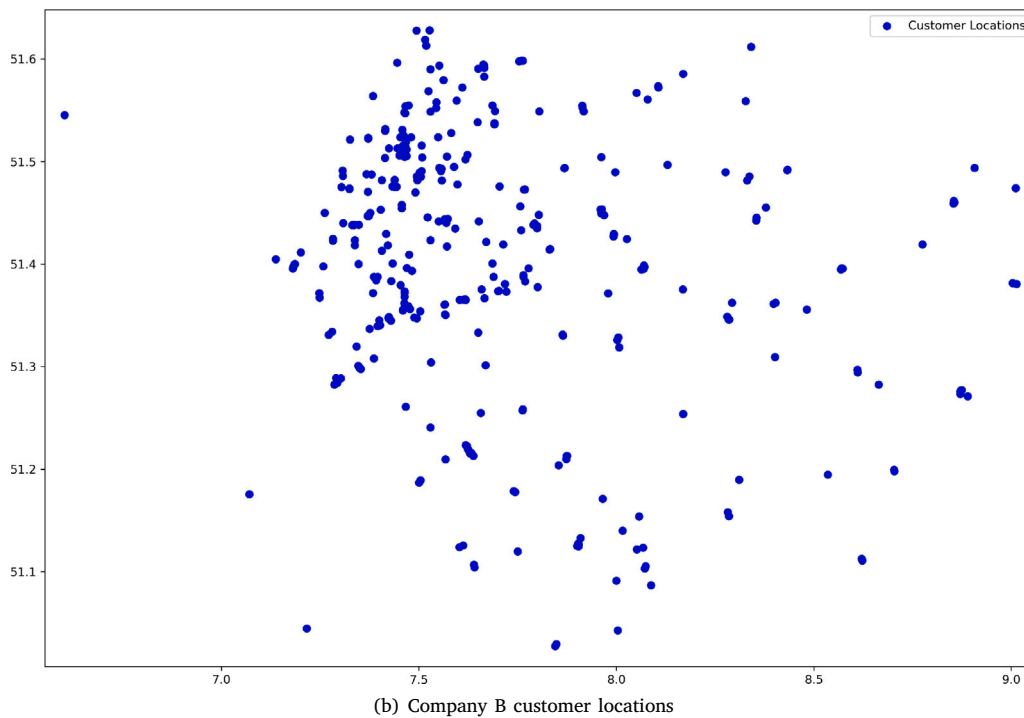
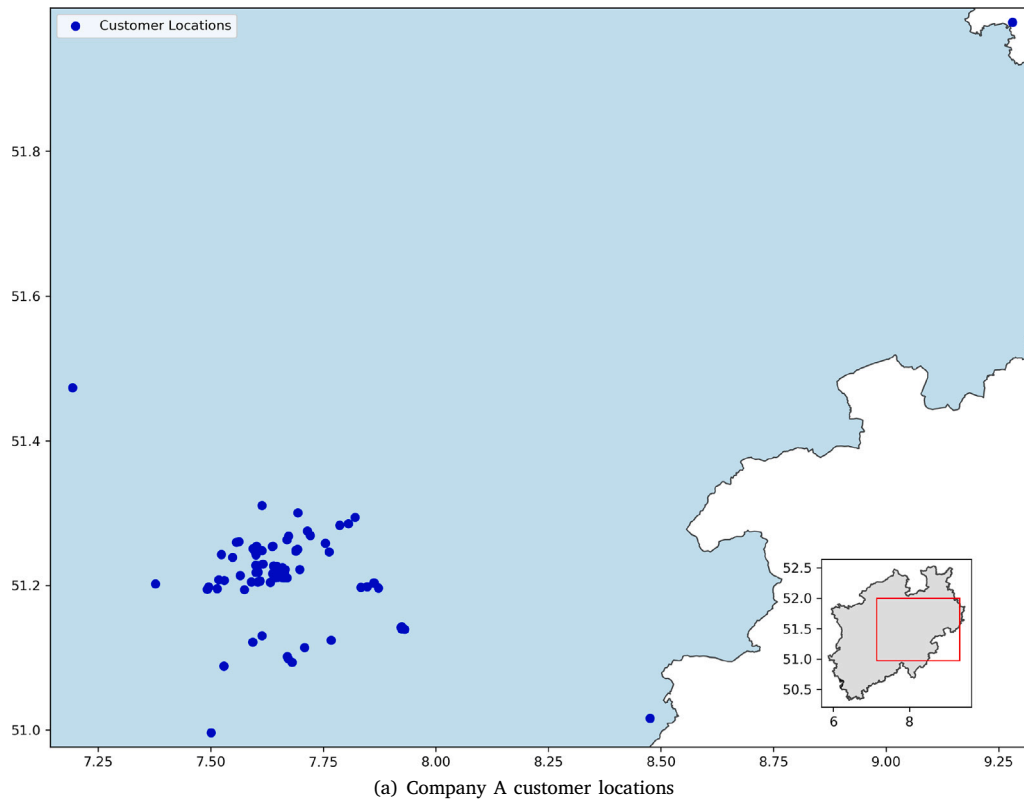


Fig. 4. Customer locations based on sample data.

and drone travel distances. By varying the coefficients (α, β) in the objective function, we examine how different operational priorities influence the allocation of distances, the number of activated micro-hubs, and the overall network configuration. This analysis allows us to distinguish between networks that remain stable under alternative weightings and those that adapt hub usage and distance distribution when truck or drone travel is emphasized. The results provide insight

into the flexibility of the model and its applicability to both compact and heterogeneous delivery systems.

As shown in Table 5, the sensitivity analysis reveals distinct behaviors across the two companies. For Company A, which represents the smaller network, the solution remains invariant across all scenarios. Drone and truck distances are identical under different (α, β) settings, and the number of activated micro-hubs is consistently six. Only the

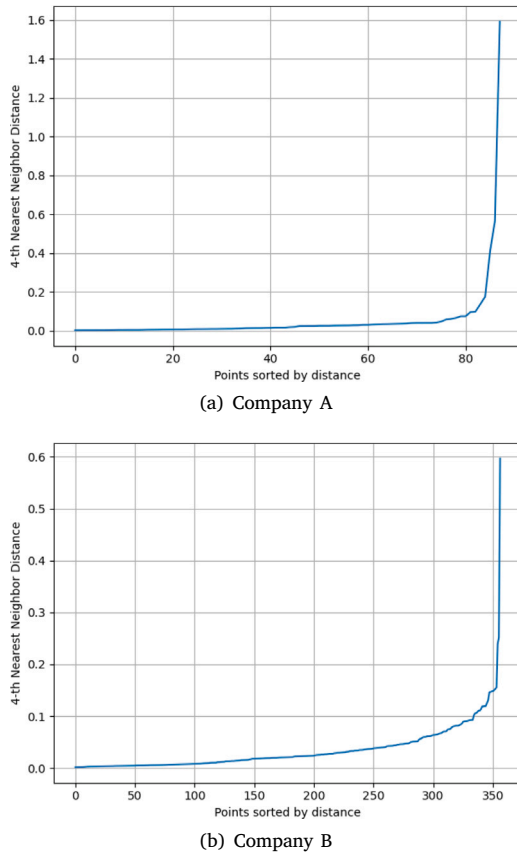


Fig. 5. Determining ϵ using the k-distance plot.

Table 4

Assignment of customer locations to the DBSCAN identified micro-hubs.

Parameter	Results	
	Company A	Company B
Drone range (R)	8 km	8 km
No. of Micro-hubs selected (DBSCAN)	6(9)	24(38)
No. of customers assigned to the Micro-hubs	67	301
No. of customers assigned to the Warehouse	15	14
Noise (Outlier)	6	42

weighted objective value changes, reflecting the rescaling of truck and drone legs. This indicates that hub placement in compact networks is robust and insensitive to relative modality weightings.

By contrast, Company B, with its larger customer base, exhibits meaningful adjustments when the weights are varied. In the baseline case ($\alpha = \beta = 1$), the solution activates 24 micro-hubs, with 997.6 km of drone travel and 807.7 km of truck travel. When truck distance is emphasized ($\alpha = 2$, $\beta = 0.5$), the structure remains stable but the weighted objective increases, reflecting the higher penalty on truck legs. Conversely, when drone distance is emphasized ($\alpha = 0.5$, $\beta = 2$), the model reduces drone travel to 886.8 km, increases truck travel to 855.6 km, and opens 26 hubs. This demonstrates that the framework adapts hub usage and distance allocation in response to operational priorities. Overall, full customer coverage is preserved in all scenarios, and the spatial topology of the network remains qualitatively stable.

These findings confirm that minimizing distance is not arbitrary. The framework distinguishes between truck and drone travel patterns, and the weighting coefficients (α, β) allow calibration to either cost or time. Practitioners may set (α, β) to modality-specific travel times (road-network times for trucks, flight times for drones) or to per-kilometer operating costs. In this way, the optimization aligns with

real-world delivery objectives: in smaller networks, solutions are robust to weighting changes, while in larger heterogeneous systems, the model adapts hub usage to reflect operational priorities.

5.2.2. Quantifying discretization loss from centroid-based micro-hub locations

For each activated micro-hub j , the snap distance $d_j^{\text{snap}} = d((x_j, y_j), (x_j^*, y_j^*))$ was computed, where (x_j^*, y_j^*) denotes the locally optimal continuous location minimizing the sum of Euclidean distances to assigned customers (see Appendix A). For Company A, the average snap distance $\bar{d}^{\text{snap}} = \frac{1}{|M^*|} \sum_{j \in M^*} d_j^{\text{snap}}$ was 0.853 km, with a maximum of 1.980 km. Relative to the drone delivery range $R = 8$ km, this corresponds to an average normalized deviation of 10.7%. Allowing local continuous refinement reduced the total drone travel distance by an average of 6.98%, with a maximum improvement of 12.22%, indicating limited discretization loss.

For Company B, the average snap distance is 0.784 km, with a maximum of 3.420 km, while the average and maximum objective improvements are 4.76% and 18.55%. These results indicate that centroid-based discretization introduces only moderate spatial displacement, while preserving solution quality within a limited and quantifiable margin. The detail is shown in Table 6.

5.3. Routing for outlier customers via intermediate hubs

The outlier customers identified by DBSCAN and left unassigned by the optimization model were serviced using the interval-based placement heuristic from Section 4.4. This method establishes a sequence of intermediate relay hubs (e.g., charging stations) along the path from the nearest activated facility to the outlier customer, ensuring each leg of the journey respects the drone's range.

Applying this heuristic, an additional 16 intermediate hubs were required to service the 6 outliers for Company A, and 47 intermediate hubs were needed for the 42 outliers of Company B. This step ensures full service coverage across the entire customer set. Since the primary objective in this study is to identify potential micro-hub locations, ranging from hubs capable of handling both packages and drones to those designed exclusively for drone charging, a comprehensive route planning of the delivery network will be addressed in future work.

To evaluate the quality of this heuristic, we benchmarked it against an exact optimization model on representative outlier instances. Specifically, for a small number of outliers (3 outliers) in each company, we compared the heuristic solution to an optimal path obtained from a flow-based MILP. The benchmark MILP is solved on a restricted geometric corridor between the facility and the customer and enforces the same operational endurance logic as the heuristic: all intermediate hops may span up to $2R$ due to recharging at relay locations, while the final hop to the customer is constrained to R , reflecting the absence of charging infrastructure at customer sites. The optimality gap is calculated as the percentage increase in distance of the MILP path over the theoretical straight-line distance utilized by the heuristic. See Appendix B for more information on benchmark formulation.

Table 7 reports the comparison results. Across all tested outliers, the heuristic produces paths that are either optimal or extremely close to optimal, with stretch values below 0.0004%. In most cases, the number of relays used by the heuristic exactly matches the optimal solution. In a small number of instances (one case from selected instances), the benchmark MILP selects more additional relay, which can be attributed to discretization effects in the corridor-based candidate grid rather than to a structural deficiency of the heuristic. In particular, the benchmark MILP restricts relay locations to a discretized corridor grid, whereas the heuristic places relays continuously along the straight-line path.

These results confirm that the proposed interval-based relay placement achieves near-optimal performance while remaining computationally efficient and scalable. This justifies its integration into the overall network planning framework, where it complements the DBSCAN-based micro-hub identification and the facility location optimization without introducing prohibitive computational overhead.

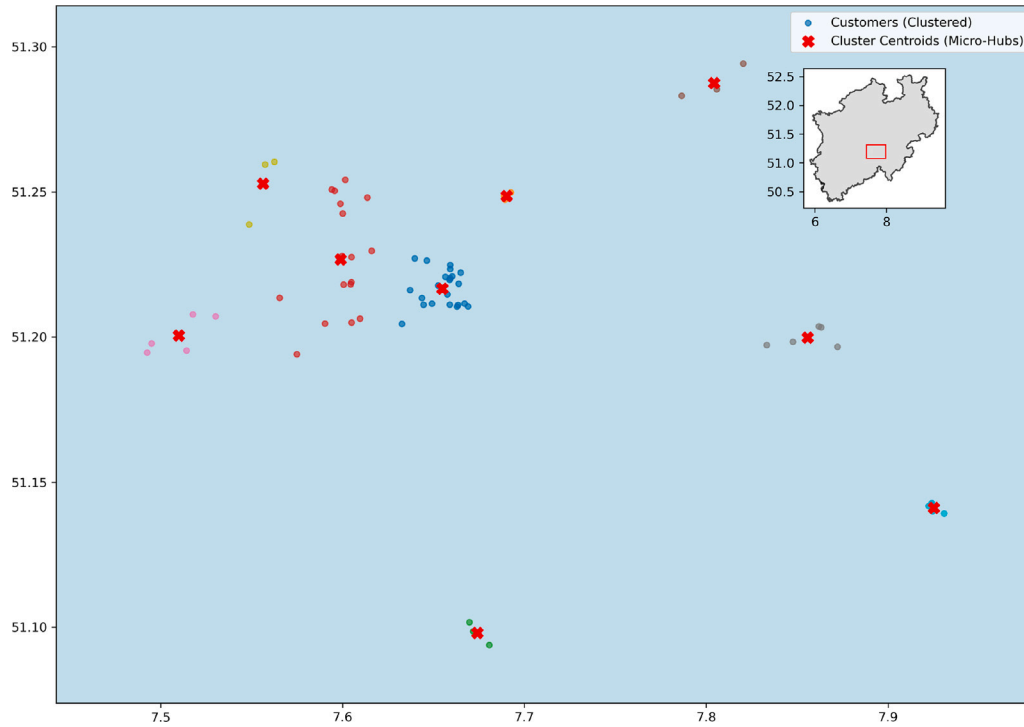
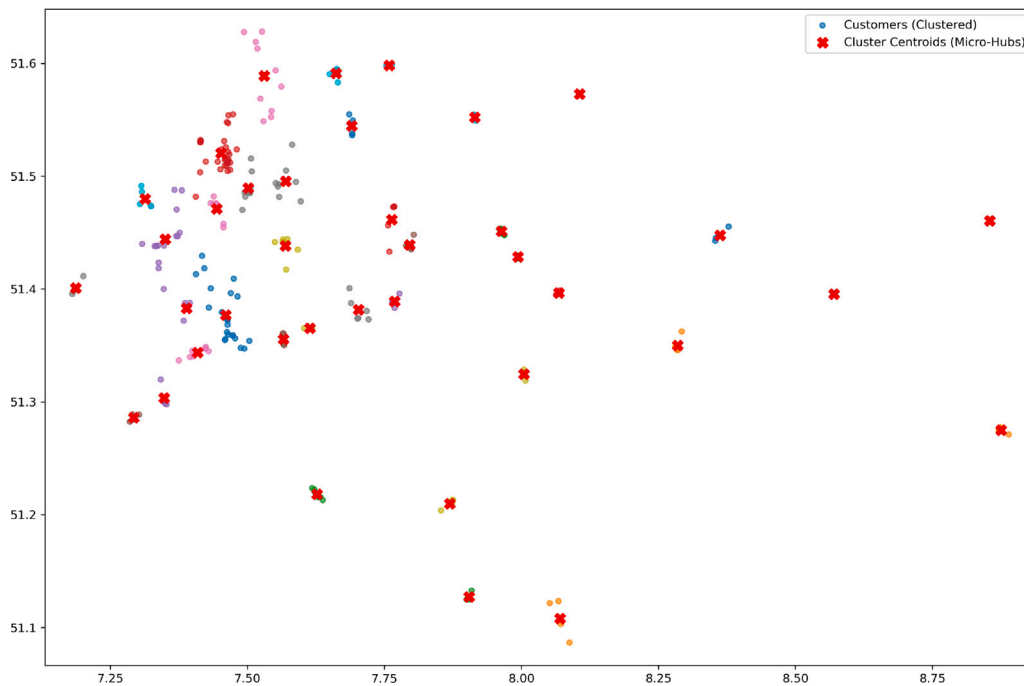
(a) Company A ($\epsilon = 0.023$, MinPts = 3)(b) Company B ($\epsilon = 0.026$, MinPts = 3)

Fig. 6. DBSCAN clustering results identifying dense customer regions and outliers.

5.4. Post-processing: Snapping hubs to feasible locations

Because the facility location model operates in continuous geographic space, some candidate micro-hub and intermediate hub locations may fall on infeasible terrain, such as water bodies, protected areas, or land parcels unsuitable for logistics operations. To ensure practical deployability, a GIS-based post-processing step is applied to relocate such facilities to nearby feasible sites.

Using detailed land-use datasets for North Rhine-Westphalia and Hessen, each candidate hub location is evaluated against a set of feasibility criteria. A location is considered feasible if it (i) lies within transport-accessible land-use categories (e.g., residential, commercial, industrial, or retail areas), (ii) does not intersect with water bodies or protected and restricted zones, and (iii) satisfies basic operational and airspace constraints. When a candidate location violates one or more of these conditions, it is snapped to the nearest feasible land parcel in

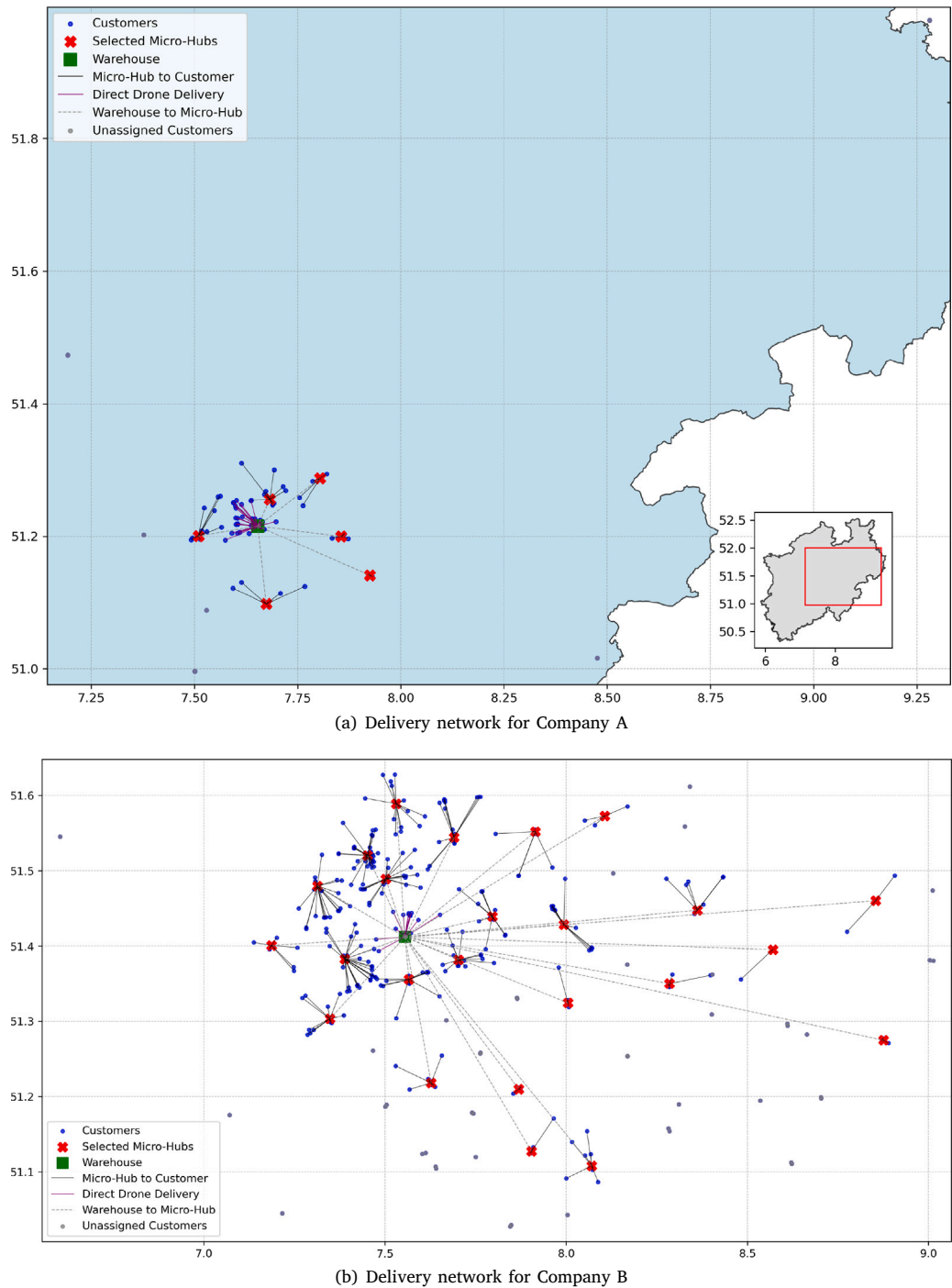


Fig. 7. Optimized delivery network showing activated micro-hubs, the warehouse, and customer assignments.

Table 5
Sensitivity to relative truck–drone distance weights.

Company	(α, β)	Distance (km)		Weighted objective	Open micro-hub
		Drone	Truck		
Company A	(1,1)	197.81	74.37	272.18	6
	(2,0.5)	197.81	74.37	247.64	6
	(0.5,2)	197.81	74.37	432.80	6
	(2,1)	197.81	74.37	346.55	6
Company B	(1,1)	997.55	807.69	1805.24	24
	(2,0.5)	997.55	807.69	2114.15	24
	(0.5,2)	886.84	855.61	2201.49	26
	(2,1)	997.55	807.69	2612.93	24

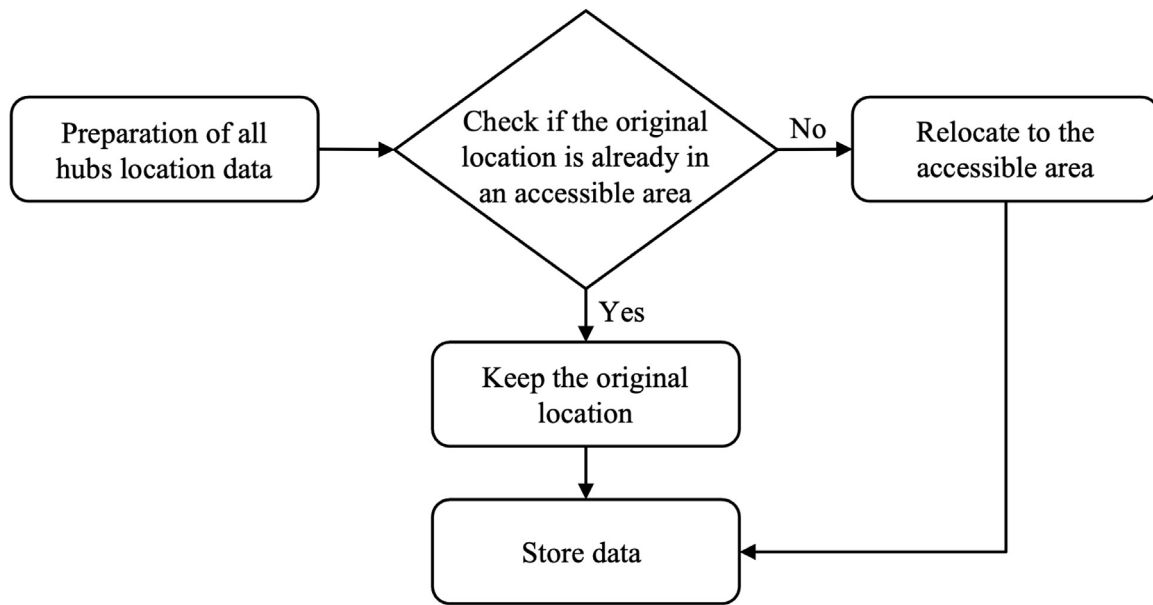


Fig. 8. Snapping process for micro-hubs and intermediate hubs to ensure placement in feasible land-use areas.

Table 6

Local refinement analysis of centroid-based micro-hubs.

Case	Snap distance (km)		Objective improvement (%)	
	Average	Maximum	Average	Maximum
Company A	0.853	1.980	6.98	12.22
Company B	0.784	3.420	4.76	18.55

Euclidean distance, thereby minimizing spatial distortion. The overall snapping procedure is illustrated in Fig. 8.

To quantify the impact of this feasibility-driven relocation, we measure the snapping displacement, defined as the distance between the original candidate location (e.g., cluster centroid or analytically derived relay position) and its snapped counterpart. Table 8 summarizes the resulting displacement statistics for both case-study companies.

Across both case studies, the resulting spatial adjustments are limited: median displacements remain well below 200 m, and more than 95% of relocated hubs are moved by less than approximately 1.1 km. Importantly, snapping does not alter customer coverage or assignment feasibility in either network. Moreover, changes in total delivery distance after relocation are marginal when compared to variations induced by key planning parameters such as drone range limits or hub activation decisions.

The snapped hub locations are subsequently treated as fixed inputs to the assignment, routing, and simulation stages. Overall, this post-processing step provides a controlled and quantifiable mechanism for bridging abstract, continuous optimization outputs with geographically and operationally feasible logistics network designs, without compromising solution structure or coverage guarantees.

5.5. Network validation and performance simulation

To assess the feasibility and performance of the designed network, a simulation was conducted using the final, snapped locations of all facilities. Dijkstra's algorithm was applied to determine the shortest delivery path for every customer. The simulation confirmed that all customers, including outliers served via intermediate hubs, could be reached within the operational constraints of the drone fleet (Fig. 9).

The simulation also quantified network efficiency in terms of total distances traveled by trucks (first echelon) and drones (second echelon). A comprehensive summary of the final network configuration

and performance metrics is reported in Table 9. These results indicate that the proposed hybrid clustering–optimization–heuristic approach is capable of designing a service-oriented and operationally feasible delivery network with full customer coverage.

It should be noted that the placement of intermediate hubs is based on a heuristic rather than an exact optimization model. While this does not guarantee global optimality, the heuristic provides a computationally tractable solution that ensures coverage of otherwise infeasible customers. This trade-off reflects a common practice in large-scale logistics planning, where heuristic solutions balance realism and tractability. A more detailed economic assessment and comparison with exact formulations (where computationally feasible) are left for future research. Here, the primary aim is to demonstrate that the framework generates complete and feasible delivery options for every customer.

5.5.1. Range sensitivity analysis

The proposed framework distinguishes between (i) a one-way operational drone range R for customer assignment and (ii) an endurance of up to $2R$ between recharge opportunities when intermediate relay hubs are available. The one-way range R corresponds to the maximum admissible distance for a loaded outbound delivery from a warehouse or micro-hub or last relay hub to a customer, ensuring sufficient battery to deliver and return back to its destination. The $2R$ endurance applies exclusively to intermediate facility-to-facility legs (warehouse–relay or micro-hub–relay, and relay–relay), where recharging is assumed to be available, and no terminal delivery is performed at the end of the leg.

To assess the robustness of feasibility and performance with respect to range assumptions, we conduct a parametric sensitivity analysis over $R \in \{0.8R_0, 0.9R_0, R_0, 1.1R_0, 1.2R_0\}$, where $R_0 = 8$ km is the baseline range used throughout the main experiments. For each value of R , the full assignment, micro-hub pruning, and relay-based routing logic is re-evaluated without altering demand, facility locations, or charging infrastructure.

Table 10 summarizes the impact of these variations on the network topology and logistics performance for both Company A and Company B.

The results demonstrate that the network design is highly robust, maintaining 100% customer coverage across all tested range values. However, the internal composition of the network shifts significantly:

Table 7

Benchmark comparison of the interval-based relay heuristic against an optimal flow-based MILP for representative outlier customers.

Company	Outlier ID	Number of relay hubs		Distance (km)		Stretch (%)
		Heuristic	Optimal	Heuristic	Optimal	
A	0	3	3	41.4256	41.4257	0.00024
	21	1	1	9.2099	9.2099	0.00000
	37	8	11	127.6507	127.6547	0.00031
B	5	1	1	16.7988	16.7988	0.00000
	7	1	1	9.2142	9.2142	0.00000
	11	1	1	14.1398	14.1398	0.00000

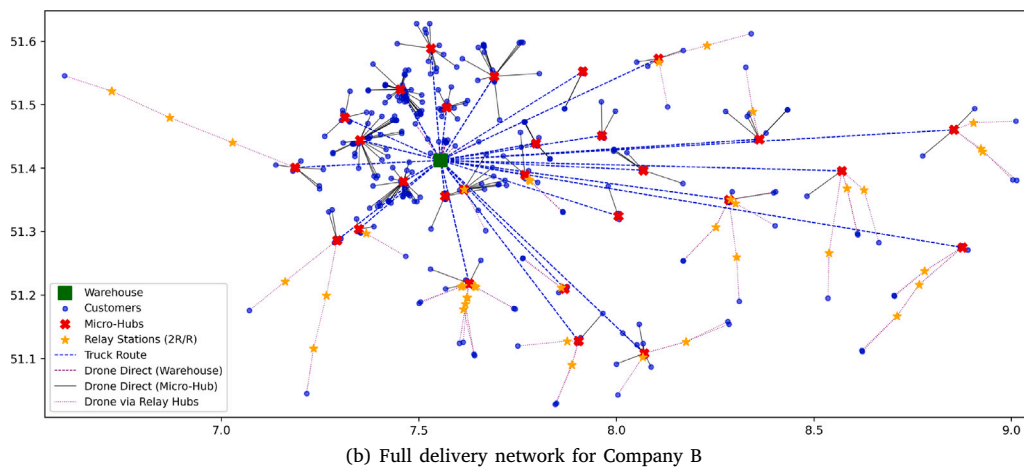
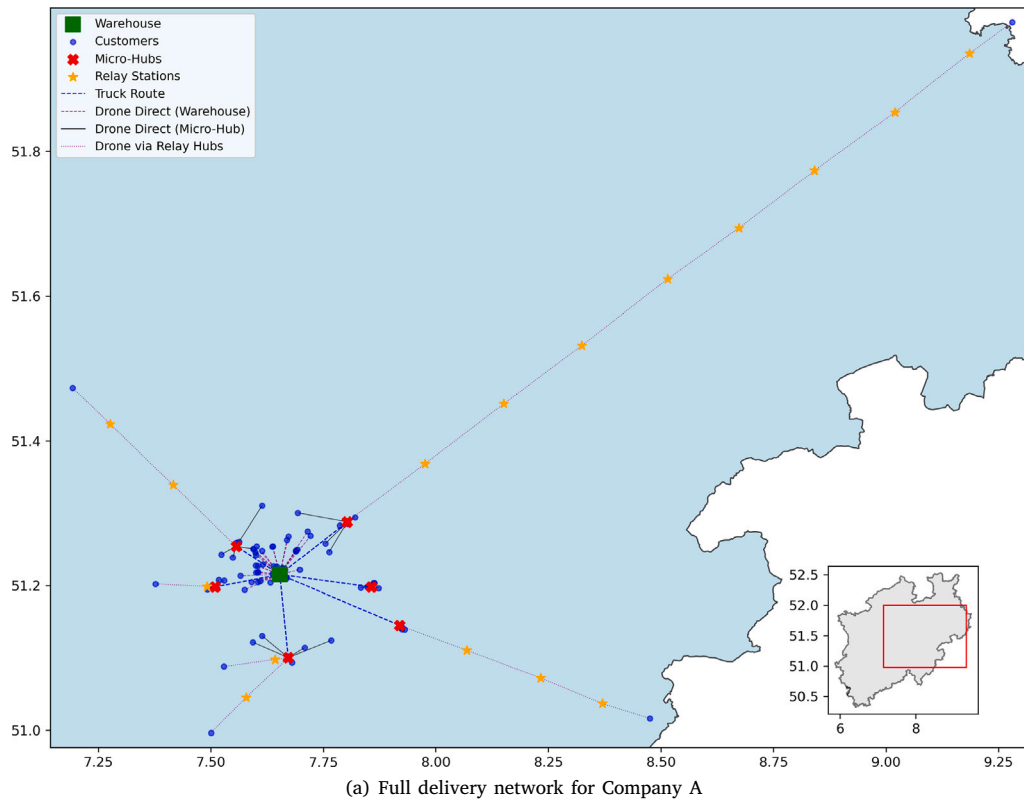


Fig. 9. Simulation results showing the complete delivery network, including routes to outlier customers via intermediate charging stations.

• **Infrastructure substitution:** As drone endurance (R) increases, the network exhibits a clear consolidation effect. The model requires fewer active micro-hubs and relay points, substituting physical infrastructure with longer aerial flight segments. This

is evidenced by the monotonic decrease in micro-hub and relay counts as R moves from 6.4 to 9.6 km.

• **Operational trade-offs:** Longer ranges enable more direct flights from the warehouse, reducing the reliance on truck-based

Table 8

Snapping displacement statistics for feasibility-driven hub relocation.

Metric	Company A	Company B
Mean displacement (km)	0.275	0.264
Median displacement (km)	0.181	0.068
95th percentile (km)	0.972	1.124
Maximum displacement (km)	1.862	1.492
Relocations > 500 m (%)	12.0	20.9
Relocations > 1 km (%)	8.0	10.5
Snapped locations (%)	96.0	75.6

micro-hub replenishment. This leads to a decrease in truck travel distance at the cost of higher total drone mileage. Conversely, at lower ranges (6.4 km), the system preserves feasibility by “densifying” the network—activating additional micro-hubs and multi-hop relay paths to bridge the gap to distant customers.

- **Outlier resilience:** The relay-based multi-hop logic proves essential at lower endurance levels. For Company B, the number of customers requiring relay assistance drops from 54 to 28 as range increases, highlighting how improved vehicle capability simplifies routing complexity by converting multi-hop paths into direct deliveries. This confirms that customer coverage at shorter ranges is not guaranteed by vehicle capability alone, but requires compensatory increases in infrastructure density through additional micro-hubs and relay hubs.

Importantly, without the activation of additional micro-hubs and relay paths, full coverage would not be attainable at lower range values. These findings confirm that the explicit distinction between R and $2R$ provides a stable and operationally realistic foundation for drone network design, enabling practitioners to trade off infrastructure density against vehicle endurance while preserving service guarantees.

5.5.2. Computational performance and scalability

To evaluate the computational scalability of the proposed framework, we report the runtime of its main algorithmic components: DBSCAN-based generation of candidate micro-hub locations, construction and solution of the assignment MILP, and the relay hub placement heuristic based on the proposed $2R/R$ endurance logic.

In addition to the original real-world datasets for Company A and Company B, two larger scenarios were generated for each case by spatially replicating customer locations with small geographic perturbations, resulting in instances with approximately $2\times$ and $4\times$ the original number of customers. For each scenario, DBSCAN clustering, MILP formulation, and relay-based feasibility restoration were executed from scratch.

All experiments were conducted on a standard workstation (MacBook M3 Pro, 18 GB RAM) using Gurobi Optimizer in a Python environment with presolve enabled and default cutting planes. The MILP corresponds to an uncapacitated assignment problem with distance-based feasibility constraints and was solved to optimality in all cases.

Table 11 reports detailed runtimes for each component. DBSCAN preprocessing consistently required only a few milliseconds, even for the largest instances. MILP solve times remained well below three seconds across all scenarios, indicating favorable scalability of the optimization core.

The relay hub placement heuristic exhibits negligible computational overhead, with execution times below 0.01 seconds per instance. This is expected, as relay locations are generated analytically along the facility–customer line using deterministic $2R/R$ rules without solving an optimization problem. Despite its negligible runtime, the heuristic plays a critical role in preserving feasibility for distant customers.

Overall, the results demonstrate that the proposed framework scales reliably from the presented datasets to substantially larger geographic regions without prohibitive runtimes, supporting its applicability for

planning-scale drone logistics network design. Importantly, the number of relay hubs grows sublinearly with customer count, reflecting the fact that relay placement is triggered only by geometric isolation rather than overall demand volume.

6. Discussion

The two case studies demonstrate that the proposed hybrid framework is both effective and practical for designing intermodal drone delivery networks. This section discusses the principal findings, interprets their implications, situates the contribution within the facility location literature, and highlights future research opportunities.

The use of cluster centroids as candidate micro-hub locations introduces an intentional discretization of the underlying continuous facility location problem. This approximation trades theoretical optimality for computational tractability and scalability, which is essential given the size and geographic resolution of the problem. Cluster centroids provide representative locations that capture spatial demand density while significantly reducing the solution space. A discretization loss analysis confirms that this approximation introduces limited spatial deviation, thereby validating centroid-based candidates as a reasonable abstraction for strategic planning.

Importantly, the proposed framework does not rely solely on centroid feasibility. Drone range constraints are explicitly enforced in the MILP, and customers that cannot be served from centroid-based locations are systematically identified as outliers. These customers are subsequently handled through a relay-based feasibility mechanism that distinguishes between a one-way operational delivery range and an extended endurance for inter-facility flights with recharging. By allowing intermediate relay hubs to bridge long distances using a structured $2R/R$ logic, discretization errors and range limitations are mitigated downstream. This ensures full service coverage while maintaining a manageable optimization structure.

The objective of the framework is strategic network design rather than exact facility placement; precise siting decisions can be refined in subsequent tactical planning stages. Similar discretization strategies are widely adopted in large-scale facility location and logistics network design to balance realism and solvability [6,55]. The present study extends this tradition by explicitly embedding feasibility recovery mechanisms within the modeling pipeline.

6.1. Findings

The results confirm the value of combining data-driven clustering with mathematical optimization. By applying DBSCAN to customer locations, the intractable continuous facility location problem is transformed into a discrete one, substantially reducing computational burden while retaining spatial flexibility. The subsequent optimization strategically activates micro-hubs in dense regions, thereby minimizing last-mile drone distances and enabling a decentralized distribution structure.

A key finding is the framework’s ability to guarantee full service coverage through the outlier-handling relay heuristic. While clustering efficiently captures dense demand, it inevitably neglects isolated customers who are operationally critical. The relay-based multi-hop logic ensures feasibility for these customers without enumerating relay locations in the MILP, preserving scalability. Sensitivity analysis further shows that reduced drone range can be compensated by activating additional micro-hubs and relay points, whereas increased range enables infrastructure consolidation, highlighting an explicit trade-off between vehicle capability and physical infrastructure.

The GIS-based snapping step further bridges theory and practice by relocating proposed facilities to viable land parcels. Displacement statistics indicate that the majority of hubs require only limited adjustment, while larger relocations remain rare and geographically constrained. This confirms that the proposed locations are not only computationally feasible but also physically realistic. Simulation results

Table 9

Final summary of network configuration and performance metrics.

Network configuration & assignment	Company A	Company B
Total Customers	88	357
<i>Customer Assignment Breakdown:</i>		
- To Micro-hubs (direct)	47	309
- To Warehouse (direct)	35	6
- To Outlier Route (via intermediate hubs)	6	42
Facilities Deployed		
Micro-hubs activated (out of candidates)	6 (9)	28 (38)
Intermediate hubs for outliers	16	47
Performance Metrics		
Total distance covered by trucks (km)	157.73	1776.10
Total distance covered by drones (km)	1005.53	3208.38

Table 10Sensitivity of feasibility and logistics metrics to drone range R (Company A and Company B).

Company	R (km)	Served	Coverage (%)	Drone (km)	Truck (km)	Customers via relays	Relay hubs	Micro-hubs
A	6.4	88	100	905.41	166.66	8	21	7
	7.2	88	100	943.42	166.66	7	18	7
	8.0	88	100	1005.53	157.73	6	16	6
	8.8	88	100	1144.27	129.07	6	14	5
	9.6	88	100	1209.79	129.07	5	13	5
B	6.4	357	100	2922.81	1871.35	54	61	32
	7.2	357	100	3101.80	1878.93	49	55	32
	8.0	357	100	3208.38	1776.10	42	47	28
	8.8	357	100	3429.38	1724.16	38	40	26
	9.6	357	100	3794.89	1587.19	28	29	22

Table 11

Computational performance under increasing instance sizes.

Company	Scenario	Customers	Micro-hubs		Outliers	Relay hubs	Time (s)		
			Candidate	Used			DBSCAN	MILP	Relay Heur.
A	Base	88	9	6	6	16	0.007	0.037	< 0.0001
	2×	176	7	6	18	40	0.001	0.057	0.0001
	4×	352	14	12	15	15	0.002	0.214	0.0001
B	Base	357	38	28	42	47	0.004	0.240	0.0006
	2×	714	47	37	76	82	0.001	1.050	0.0013
	4×	1428	62	45	220	220	0.002	2.240	0.0046

validate that the integrated network – including micro-hubs, relay hubs, and warehouse connections – can serve the complete customer base within operational constraints.

6.2. Implications and contribution

Practical implications: The framework provides logistics operators with a scalable decision-support tool for strategic hub planning. It enables investment in micro-hubs where customer density justifies them while systematically covering sparse regions through relay-assisted drone operations. The explicit distinction between operational delivery range and inter-facility endurance allows operators to assess how vehicle capability, infrastructure density, and service guarantees interact. By incorporating GIS-based feasibility checks, the approach reduces the risk of infeasible site choices and enhances the credibility of drone logistics deployment in mixed urban–rural environments.

Contribution to the literature: This study contributes to the facility location and drone logistics literature by integrating clustering, optimization, relay-based feasibility recovery, and spatial validation in a single framework. While clustering-based approaches such as Han and Liu [54] have been explored, they generally lack mechanisms to ensure coverage of outlier demand. Similarly, studies on drone recharging infrastructure (e.g., Hong et al. [50]; Shavarani et al. [51]) typically assume predefined candidate sites. By contrast, the proposed framework determines both micro-hub and relay hub locations dynamically in continuous space, then validates their feasibility through GIS snapping, sensitivity analysis, and simulation. This integration extends

the facility location tradition in logistics and transportation [e.g., 11] while addressing operational constraints unique to drone-based delivery networks.

6.3. Limitations and future work

The present study assumes equal customer weights, ignores capacity constraints at micro-hubs, and abstracts intermediate hub placement through a heuristic mechanism. While this ensures tractability and scalability, it does not guarantee global optimality. Moreover, the model focuses exclusively on strategic location decisions and does not explicitly incorporate operational elements such as routing, fleet scheduling, workload balancing, or time-dependent effects. As a result, strategically efficient hub configurations may still lead to operational challenges, including temporary bottlenecks, uneven vehicle utilization, increased waiting times, or the need for additional fleet resources under peak demand conditions.

These simplifications are appropriate for early-stage strategic planning but necessitate subsequent operational validation before real-world implementation. Extensions could incorporate heterogeneous demand, hub capacities, explicit cost structures, and uncertainty through robust or stochastic formulations. Further integration of routing optimization, temporal constraints, and sustainability objectives would strengthen the operational relevance of the framework and support practical deployment.

7. Conclusions

This study presented a hybrid clustering–optimization framework for identifying micro-hub and intermediate hub locations in intermodal drone delivery networks. By combining DBSCAN clustering with a facility location model in continuous space, the approach transforms an otherwise intractable problem into a scalable and computationally manageable one. A relay-based feasibility heuristic is introduced to systematically handle customers beyond direct drone range, ensuring complete service coverage under realistic endurance constraints. The integration of GIS-based snapping and network simulation further strengthens the practical applicability of the proposed framework.

Application to real-world datasets from Germany demonstrates that the framework produces deployable network designs that balance efficiency and coverage across both dense urban areas and sparsely populated regions. Sensitivity analysis confirms that full coverage can be maintained across a wide range of drone capabilities by adjusting the density of micro-hubs and relay hubs, thereby revealing a clear trade-off between vehicle endurance and infrastructure intensity. Scalability experiments further show that the combined DBSCAN–MILP pipeline remains computationally tractable as instance size increases, supporting its suitability for large regional planning problems.

The main contribution of this work lies in demonstrating how data-driven clustering, mathematical optimization, feasibility recovery mechanisms, and spatial validation can be integrated into a single decision-support tool for strategic drone logistics planning. Rather than aiming for exact facility placement, the framework provides robust and interpretable network configurations that can guide investment and policy decisions in early planning stages. While the current model assumes equal demand, omits capacity constraints, and treats relay placement heuristically, these simplifications are appropriate for strategic analysis and enable scalability to real-world problem sizes.

Future research should extend the framework by incorporating heterogeneous demand, hub capacity constraints, explicit cost and emission models, and uncertainty through robust or stochastic formulations. Integrating detailed routing decisions, temporal considerations, and sustainability objectives would further enhance operational relevance and support the deployment of resilient and efficient drone-assisted logistics networks.

CRedit authorship contribution statement

Asrat Gobachew: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Diyar Altinses:** Resources, Formal analysis. **Andreas Schwung:** Writing – review & editing, Resources, Project administration, Methodology, Funding acquisition. **Stefan Lier:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Quantifying discretization loss from centroid-based micro-hub locations

To quantify the loss introduced by centroid-based discretization of micro-hub locations, we conducted a local continuous refinement analysis on the set of activated micro-hubs. This analysis isolates the approximation error arising from representing micro-hub locations by discrete cluster centroids, independent of any GIS-based feasibility relocation.

Let M denote the set of candidate micro-hubs obtained via DBSCAN clustering, and let $y_j \in \{0, 1\}$ indicate whether micro-hub $j \in M$ is activated in the integrated optimization model. For each activated micro-hub j with $y_j = 1$, we define the set of customers assigned to that hub as

$$C_j = \{k \in C \mid \text{customer } k \text{ is assigned to hub } j\},$$

where each customer $k \in C_j$ has geographic coordinates (x_k, y_k) .

The discrete micro-hub location (x_j, y_j) corresponds to the DBSCAN cluster centroid used in the optimization model. To assess the best achievable continuous location for the same assignment, we compute a locally optimal refined position (x_j^*, y_j^*) by solving the following continuous Weber problem:

$$(x_j^*, y_j^*) = \arg \min_{(x, y) \in \mathbb{R}^2} \sum_{k \in C_j} d((x, y), (x_k, y_k)),$$

where $d(\cdot, \cdot)$ denotes the Euclidean distance. This refinement represents the best possible continuous micro-hub location given fixed customer assignments.

The snap distance for micro-hub j is then defined as the distance between the centroid-based discrete location and its locally optimal continuous refinement:

$$d_j^{\text{snap}} = d((x_j, y_j), (x_j^*, y_j^*)).$$

To summarize the discretization loss across the network, we report the average and maximum snap distances over all activated micro-hubs:

$$\bar{d}^{\text{snap}} = \frac{1}{|M^*|} \sum_{j \in M^*} d_j^{\text{snap}}, \quad d_{\max}^{\text{snap}} = \max_{j \in M^*} d_j^{\text{snap}},$$

where $M^* = \{j \in M \mid y_j = 1\}$.

In addition to spatial displacement, we evaluate the impact of discretization on the objective value by recomputing total delivery distances using the refined locations (x_j^*, y_j^*) while keeping customer assignments fixed. The relative increase in total distance incurred by centroid-based discretization is reported as

$$\Delta D^{\text{disc}} = \frac{D^{\text{centroid}} - D^{\text{refined}}}{D^{\text{refined}}} \times 100\%,$$

where D^{centroid} and D^{refined} denote the total delivery distance under centroid-based and refined micro-hub locations, respectively.

This analysis provides an upper bound on the loss induced by spatial discretization and demonstrates that centroid-based representations introduce only limited deviation from locally optimal continuous placements.

Algorithm 3 Snap Distance Computation via Local Continuous Refinement

- 1: **for all** activated micro-hubs j **do**
- 2: Retrieve assigned customer set C_j
- 3: Compute centroid location (x_j, y_j)
- 4: Compute local continuous location (x_j^*, y_j^*) by minimizing $\sum_{k \in C_j} d((x, y), (x_k, y_k))$
- 5: Compute snap distance $d_j^{\text{snap}} \leftarrow \|(x_j, y_j) - (x_j^*, y_j^*)\|$
- 6: **end for**
- 7: Compute summary statistics: average and maximum snap distance

Appendix B. Benchmarking the interval-based relay heuristic

The interval-based relay placement heuristic is designed to restore feasibility for customers located beyond the one-way drone range R by introducing intermediate relay hubs, while preserving near-optimal path length. This appendix provides theoretical bounds and an empirical optimality benchmark for this heuristic.

Relay feasibility rule. Let f denote a facility (warehouse or micro-hub), k a customer location, and r an intermediate relay hub. The heuristic enforces the following endurance constraints:

$$d_{uv} \leq \begin{cases} R, & v = k, \\ 2R, & \text{otherwise,} \end{cases} \quad (\text{B.1})$$

where d_{uv} denotes the geodesic distance between consecutive nodes u and v . The tighter bound on arcs terminating at customers reflects the absence of charging infrastructure at customer locations, whereas intermediate facilities are assumed to allow recharging.

Bound on the number of relays. For a customer at distance $D = d_{fk}$ from its serving facility, feasibility under rule (B.1) is ensured by introducing at most

$$n_{\text{hubs}} = \max \left\{ 0, \left\lceil \frac{D - R}{2R} \right\rceil \right\} \quad (\text{B.2})$$

relay hubs, implying linear growth of relay count with distance and no exponential complexity.

Optimality benchmark formulation. To assess the quality of the heuristic, we benchmark it against a shortest-path mixed-integer linear programming (MILP) formulation defined on a discretized corridor between f and k . Let $\mathcal{V} = \{f, k\} \cup \mathcal{R}$ denote the node set, where $\mathcal{R}$ is a set of candidate relay locations sampled along the straight-line corridor between f and k . Binary decision variables x_{uv} indicate whether arc (u, v) is used and $\mathcal{A}$ is set of feasible directed arcs between nodes.

The benchmark model is:

$$\min \sum_{(u,v) \in \mathcal{A}} d_{uv} x_{uv} \quad (\text{B.3})$$

$$\text{s.t.} \sum_{v: (f,v) \in \mathcal{A}} x_{fv} = 1, \quad (\text{B.4})$$

$$\sum_{u: (u,k) \in \mathcal{A}} x_{uk} = 1, \quad (\text{B.5})$$

$$\sum_{u: (u,v) \in \mathcal{A}} x_{uv} = \sum_{w: (v,w) \in \mathcal{A}} x_{vw}, \quad \forall v \in \mathcal{R}, \quad (\text{B.6})$$

$$d_{uv} \leq R \quad \text{if } v = k, \quad (\text{B.7})$$

$$d_{uv} \leq 2R \quad \text{otherwise,} \quad (\text{B.8})$$

$$x_{uv} \in \{0, 1\}, \quad \forall (u, v) \in \mathcal{A}. \quad (\text{B.9})$$

Benchmark results. The heuristic is evaluated against the MILP on a subset of outlier customers (three per company). Across all instances, the heuristic produces paths that are either identical to or within 0.0003% of the optimal MILP solution. Differences are on the order of meters over tens of kilometers and are attributable to numerical discretization and floating-point geometry rather than structural suboptimality. Importantly, the MILP never identifies a path with a different topology or a strictly smaller relay count than the heuristic.

We report the relative stretch between the proposed relay heuristic and a benchmark mixed-integer linear program (MILP) solved on a discretized relay candidate set. The stretch is defined as

$$\text{Stretch}(\%) = 100 \times \frac{d^{\text{MILP}} - d^{\text{heur}}}{d^{\text{heur}}},$$

where d^{heur} denotes the total length of the continuous heuristic path and d^{MILP} denotes the optimal path length obtained by the discretized benchmark MILP.

Because the heuristic places relay points in continuous space, while the MILP is restricted to a finite candidate set, the MILP provides a conservative upper bound on the true optimum. Consequently, small positive stretch values reflect discretization effects rather than suboptimality of the heuristic.

Data availability

We can provide a synthetic dataset closely matching the statistical and spatial properties of the case studies, and the full implementation code.

References

- [1] Feng X, Murray AT, Church RL. Drone service response: Spatiotemporal heterogeneity implications. *J Transp Geogr* 2021;93(September 2020):103074. <http://dx.doi.org/10.1016/j.jtrangeo.2021.103074>.
- [2] Lin M, Lyu JY, Gao JJ, Li LY. Model and hybrid algorithm of collaborative distribution system with multiple drones and a truck. *Sci Program* 2020;2020. <http://dx.doi.org/10.1155/2020/8887057>.
- [3] Moadab A, Farajzadeh F, Fatahi Valilai O. Drone routing problem model for last-mile delivery using the public transportation capacity as moving charging stations. *Sci Rep* 2022;12(1):1–16. <http://dx.doi.org/10.1038/s41598-022-10408-4>.
- [4] Ogazón E, Anaya-Arenas AM, Ruiz A. Designing hub-based regional transportation networks with service level constraints. *Transp Res Part E: Logist Transp Rev* 2025;195(June 2024). <http://dx.doi.org/10.1016/j.tre.2025.103991>.
- [5] Gobachew A, Haasis H-D. Scenario-based optimization of supply chain performance under demand uncertainty. *Sustain (Switzerland)* 2023;15(13). <http://dx.doi.org/10.3390/su151310603>.
- [6] Daskin MS. *Network and discrete location: models, algorithms, and applications*. 2nd ed.. Hoboken, New Jersey: John Wiley & Sons, Inc.; 2013, LK - <http://worldcat.org/title/841039606>. URL <http://site.ebrary.com/id/10726741>.
- [7] Gao X, Park C, Chen X, Xie E, Huang G, Zhang D. Globally optimal facility locations for continuous-space facility location problems. *Appl Sci (Switzerland)* 2021;11(16). <http://dx.doi.org/10.3390/app11167321>.
- [8] Drezner Z. Continuous facility location problem. *The palgrave handbook of operations research*. Cham: Springer International Publishing; 2022, p. 269–306. http://dx.doi.org/10.1007/978-3-030-96935-6_9.
- [9] Moshref-Javadi M, Winkenbach M. Applications and research avenues for drone-based models in logistics: A classification and review. *Expert Syst Appl* 2021;177(December 2019):114854. <http://dx.doi.org/10.1016/j.eswa.2021.114854>.
- [10] Garg V, Niranjana S, Prybutok V, Pohlen T, Gligor D. Drones in last-mile delivery: A systematic review on efficiency, accessibility, and sustainability. *Transp Res Part D: Transp Environ* 2023;123(April):103831. <http://dx.doi.org/10.1016/j.trd.2023.103831>.
- [11] Saldanha-da Gama F. Facility location in logistics and transportation: An enduring relationship. *Transp Res Part E: Logist Transp Rev* 2022;166(July):102903. <http://dx.doi.org/10.1016/j.tre.2022.102903>.
- [12] Farahani RZ, Hekmatfar M. *Facility Location: concepts, models, algorithms and case studies*. 1st ed. Berlin, Heidelberg: Springer; 2009, p. 464. [http://www.icclab.nl/fileadmin/default/content/erim/research/centres/erasmus_centre_for_cooperatives\(ecc\)/research/articles/c22007orientationindiversificationbehaviorofcoope.pdf](http://www.icclab.nl/fileadmin/default/content/erim/research/centres/erasmus_centre_for_cooperatives(ecc)/research/articles/c22007orientationindiversificationbehaviorofcoope.pdf).
- [13] Hakimi SL. Optimum locations of switching centers and the absolute centers and medians of a graph. *Oper Res* 1964;12(3):450–9. <http://dx.doi.org/10.1287/opre.12.3.450>.
- [14] Cooper L. Location-allocation problems. *Oper Res* 1963;11(3):331–43. http://dx.doi.org/10.1007/978-3-7908-2151-2_5.
- [15] Zhao R, Xiao Y, Luo R, Yang R, Zhou S, Zhang S. Discrete-continuous model for facility location problem with capacity-cost relation constraints. *Comput Ind Eng* 2023;185(February 2022):109661. <http://dx.doi.org/10.1016/j.cie.2023.109661>.
- [16] Wang W, Wu S, Wang S, Zhen L, Qu X. Emergency facility location problems in logistics: Status and perspectives. *Transp Res Part E: Logist Transp Rev* 2021;154(January):102465. <http://dx.doi.org/10.1016/j.tre.2021.102465>.
- [17] Church R, ReVelle C. The maximal covering location problem. *Pap Reg Sci Assoc* 1974;32(1):101–18. <http://dx.doi.org/10.1007/BF01942293>, URL <http://link.springer.com/10.1007/BF01942293>.
- [18] Melo M, Nickel S, Saldanha-da Gama F. Facility location and supply chain management – A review. *European J Oper Res* 2009;196(2):401–12. <http://dx.doi.org/10.1016/j.ejor.2008.05.007>.
- [19] Weber A. In: Friedrich CJ, editor. *Über den Standort der Industrien*. English Tr ed. Chicago: University of Chicago Press; 1909.
- [20] Hekmatfar M, Pishvaei M. Facility location: Concepts, models, algorithms and case studies. In: *Management science*. Physica-Verlag Heidelberg; 2009, p. 243–70. <http://dx.doi.org/10.1007/978-3-7908-2151-2>, URL <http://link.springer.com/10.1007/978-3-7908-2151-2>.

- [21] Love RF, Morris JG, Wesolowsky GO. Facility location: Models and methods publication in operations research. Elsevier Science Publishing Co., Inc.; 1988.
- [22] Brimberg J, Kakhki HT, Wesolowsky GO. Location among regions with varying norms. *Ann Oper Res* 2003;122(1–4):87–102. <http://dx.doi.org/10.1023/A:1026190322164>.
- [23] Rosing KE, ReVelle CS. Heuristic concentration: Two stage solution construction. *European J Oper Res* 1997;97(1):75–86. [http://dx.doi.org/10.1016/S0377-2217\(96\)00100-2](http://dx.doi.org/10.1016/S0377-2217(96)00100-2).
- [24] Snyder LV. Facility location under uncertainty: A review. *IIE Trans (Inst Ind Eng)* 2006;38(7):547–64. <http://dx.doi.org/10.1080/07408170500216480>.
- [25] Owen SH, Daskin MS. Strategic facility location: A review. *European J Oper Res* 1998;111(3):423–47. [http://dx.doi.org/10.1016/S0377-2217\(98\)00186-6](http://dx.doi.org/10.1016/S0377-2217(98)00186-6).
- [26] Farahani RZ, Hekmatfar M, Arabani AB, Nikbaksh E. Hub location problems: A review of models, classification, solution techniques, and applications. *Comput Ind Eng* 2013;64(4):1096–109. <http://dx.doi.org/10.1016/j.cie.2013.01.012>.
- [27] Laporte G. The traveling salesman problem: An overview of exact and approximate algorithms. *European J Oper Res* 1992;59(2):231–47. [http://dx.doi.org/10.1016/0377-2217\(92\)90138-Y](http://dx.doi.org/10.1016/0377-2217(92)90138-Y).
- [28] Resende MG, Werneck RF. A hybrid multistart heuristic for the uncapacitated facility location problem. *European J Oper Res* 2006;174(1):54–68. <http://dx.doi.org/10.1016/j.ejor.2005.02.046>.
- [29] Meira LA, Miyazawa FK, Pedrosa LL. Clustering through continuous facility location problems. *Theoret Comput Sci* 2017;657:137–45. <http://dx.doi.org/10.1016/j.tcs.2016.10.001>.
- [30] Gupta A, Tangwongsan K. Simpler analyses of local search algorithms for facility location. 2008, p. 1–17, URL <http://arxiv.org/abs/0809.2554>.
- [31] Dukkanci O, Campbell JF, Kara BY. Facility location decisions for drone delivery with riding: A literature review. *Comput Oper Res* 2024;167(February 2023):397–418. <http://dx.doi.org/10.1016/j.cor.2024.106672>.
- [32] Kumar K, Kumanan S. Decision making in location selection: An integrated approach with clustering and TOPSIS non traditional manufacturing view project my Ph.D work view project. *IUP* 2012;XI(1):7–21, URL <https://www.researchgate.net/publication/255726019>.
- [33] Al-Yakoob SM, Sherali HD. A mathematical modelling and optimization approach for a maritime facility location transshipment problem. *Inform (Netherlands)* 2018;29(4):609–32. <http://dx.doi.org/10.15388/Informatica.2018.184>.
- [34] Ester M, Kriegel HP, Sander J, Xu X. A density-based algorithm for discovering clusters in large spatial databases with noise. In: *Second international conference on knowledge discovery and data mining*, vol. KDD'96, 1996, p. 226–31. <http://dx.doi.org/10.1016/B978-0-444-64165-6.03005-6>.
- [35] Lyu Z, Pons D, Palliparampil G, Zhang Y. Optimising urban freight logistics using discrete-event simulation and cluster analysis: A stochastic two-tier hub-and-spoke architecture approach. *Smart Cities* 2023;6(5):2347–66. <http://dx.doi.org/10.3390/smartcities6050107>.
- [36] Zhang W, Reimann M. Towards a multi-objective performance assessment and optimization model of a two-echelon supply chain using SCOR metrics. *Central Eur J Oper Res* 2014;22(4):591–622. <http://dx.doi.org/10.1007/s10100-013-0294-7>.
- [37] De Felice F, Petrillo A. Hierarchical model to optimize performance in logistics policies: Multi attribute analysis. *Procedia - Soc Behav Sci* 2012;58:1555–64. <http://dx.doi.org/10.1016/j.sbspro.2012.09.1142>.
- [38] Barnhart C, Clausen U, Lauther U, Möhring R. Models and algorithms for optimization in logistics. *Gagstuhl Semin* 09261 2009;1–2, URL <http://drops.dagstuhl.de/opus/volltexte/2009/2175>.
- [39] Lin T, Liu Y, Liu B, Wang Y, Wu S, Zhe W. Hierarchical clustering framework for facility location selection with practical constraints. *IET Cyber-Phys Syst: Theory Appl* 2021;6(4):238–53. <http://dx.doi.org/10.1049/cps2.12021>.
- [40] Alvarez-Urbe KC, Gañan-Cardenas E, Perez-Montoya D. Hybrid clustering strategy for micro-hubs location in newspaper distribution. *Oper Supply Chain Manag* 2023;16(4):473–87. <http://dx.doi.org/10.31387/oscm0550405>.
- [41] Wu J, Liu X, Li Y, Yang L, Yuan W, Ba Y. A two-stage model with an improved clustering algorithm for a distribution center location problem under uncertainty. *Mathematics* 2022;10(14). <http://dx.doi.org/10.3390/math10142519>.
- [42] Sharma A, Sharma A, Jalal A, Kant K. A two step clustering method for facility location problem. *Int J Adv Intell Parad* 2018;10(1):1. <http://dx.doi.org/10.1504/ijaip.2018.10011478>.
- [43] Sharma A, Sharma A, Jalal AS. Clustering based hybrid approach for facility location problem. *Manag Sci Lett* 2017;7(12):577–84. <http://dx.doi.org/10.5267/j.msl.2017.8.007>.
- [44] Jazairy A, Persson E, Brho M, von Haartman R, Hilletoft P. Drones in last-mile delivery: a systematic literature review from a logistics management perspective. *Int J Logist Manag* 2024;36(7):1–62. <http://dx.doi.org/10.1108/IJLM-04-2023-0149>.
- [45] Altines D, Torres DOS, Gobachew AM, Lier S, Schwung A. Synthetic dataset generation for optimizing multimodal drone delivery systems. *Drones* 2024;8(12):1–27. <http://dx.doi.org/10.3390/drones8120724>.
- [46] Benarbia T, Kyamakya K. A literature review of drone-based package delivery logistics systems and their implementation feasibility. *Sustain (Switzerland)* 2022;14(1). <http://dx.doi.org/10.3390/su14010360>.
- [47] Pratelli A, Brocchini L, Petri M. Optimising drone-assisted logistics for urban last-mile delivery: An overview of applications, methodologies, and emerging trends. *J Urban Dev Manag* 2024;3(3):188–97.
- [48] Rodríguez-García MÁ, Fernández A, Billhardt H. Dynamic delivery plan adaptation in open systems. In: *Lecture notes in computer science (including subseries lecture notes in artificial intelligence and lecture notes in bioinformatics)*, vol. 11327 LNAI, 2019, p. 190–8. http://dx.doi.org/10.1007/978-3-030-17294-7_14.
- [49] Wang X, Zhao J, Cheng C, Qi M. A multi-objective fuzzy facility location problem with congestion and priority for drone-based emergency deliveries. *Comput Ind Eng* 2023;179(March):109167. <http://dx.doi.org/10.1016/j.cie.2023.109167>.
- [50] Hong I, Kuby M, Murray AT. A range-restricted recharging station coverage model for drone delivery service planning. *Transp Res Part C: Emerg Technol* 2018;90(February):198–212. <http://dx.doi.org/10.1016/j.trc.2018.02.017>.
- [51] Shavarani SM, Nejad MG, Rismanchian F, Izbirak G. Application of hierarchical facility location problem for optimization of a drone delivery system: a case study of Amazon prime air in the city of San Francisco. *Int J Adv Manuf Technol* 2018;95(9–12):3141–53. <http://dx.doi.org/10.1007/s00170-017-1363-1>.
- [52] Shavarani SM, Mosallaeipour S, Golabi M, Izbirak G. A congested capacitated multi-level fuzzy facility location problem: An efficient drone delivery system. *Comput Oper Res* 2019;108:57–68. <http://dx.doi.org/10.1016/j.cor.2019.04.001>.
- [53] Contreras I, Cordeau JF, Laporte G. Benders decomposition for large-scale uncapacitated hub location. *Oper Res* 2011;59(6):1477–90.
- [54] Han X, Liu X. Equilibrium decisions for multi-firms considering consumer quality preference. *Int J Prod Econ* 2020;227(July 2019):107688. <http://dx.doi.org/10.1016/j.ijpe.2020.107688>.
- [55] Chakrabarty D, Negahbani M, Sarkar A. Approximation algorithms for continuous clustering and facility location problems. In: *Leibniz international proceedings in informatics, LIPIcs*, vol. 244, no. 33, Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl Publishing, Germany; 2022, p. 33:1–33:15. <http://dx.doi.org/10.4230/LIPIcs.EA.2022.33>.
- [56] Farahani RZ, SteadieSeifi M, Asgari N. Multiple criteria facility location problems: A survey. *Appl Math Model* 2010;34(7):1689–709. <http://dx.doi.org/10.1016/j.apm.2009.10.005>.
- [57] Kim K, Moon S, Lee J, Kim J. Environmental impacts of transportation policy and vehicle market share predictions. *Transp Res Part D: Transp Environ* 2023;114(November 2022):103550. <http://dx.doi.org/10.1016/j.trd.2022.103550>.
- [58] Kaufman L, Rousseeuw PJ. Finding groups in data: an introduction to cluster analysis. John Wiley & Sons, Inc.; 1990, p. 6.
- [59] Hahsler M, Piekenbrock M, Doran D. DbSCAN: Fast density-based clustering with R. *J Stat Softw* 2019;91(1). <http://dx.doi.org/10.18637/jss.v091.i01>.
- [60] Zhang M. Use density-based spatial clustering of applications with noise (DBSCAN) algorithm to identify galaxy cluster members. *IOP Conf Ser: Earth Environ Sci* 2019;252(4). <http://dx.doi.org/10.1088/1755-1315/252/4/042033>.
- [61] Li M, Meng D, Gu S, Liu S. Research and improvement of DBSCAN cluster algorithm. In: *Proceedings - 2015 7th international conference on information technology in medicine and education*. 2016, p. 537–40. <http://dx.doi.org/10.1109/ITME.2015.100>.
- [62] Eilers PH, Marx BD. Flexible smoothing with B-splines and penalties. *Statist Sci* 1996;11(2):89–102. <http://dx.doi.org/10.1214/ss/1038425655>.
- [63] Hamed Javadi S, Guerrero A, Mouazen AM. Clustering and smoothing pipeline for management zone delineation using proximal and remote sensing. *Sensors* 2022;22(2). <http://dx.doi.org/10.3390/s22020645>.
- [64] Freitas JC, Penna PHV, Toffolo TA. Exact and heuristic approaches to truck-drone delivery problems. *EURO J Transp Logist* 2023;12(July 2022):100094. <http://dx.doi.org/10.1016/j.ejtl.2022.100094>.
- [65] Schermer D, Moeini M, Wendt O. Algorithms for solving the vehicle routing problem with drones. *Lecture Notes in Artificial Intelligence* 2018;48–58. <http://dx.doi.org/10.1007/978-3-319-75417-8>.